

The background of the slide is black, featuring a complex, abstract pattern of thin, glowing lines in shades of purple and white. These lines are mostly concentrated in the upper half of the image, creating a sense of depth and movement, resembling a network or a quantum field. The lines vary in thickness and brightness, with some appearing as sharp, straight paths and others as more diffuse, swirling clouds.

超導量子計算是什麼？

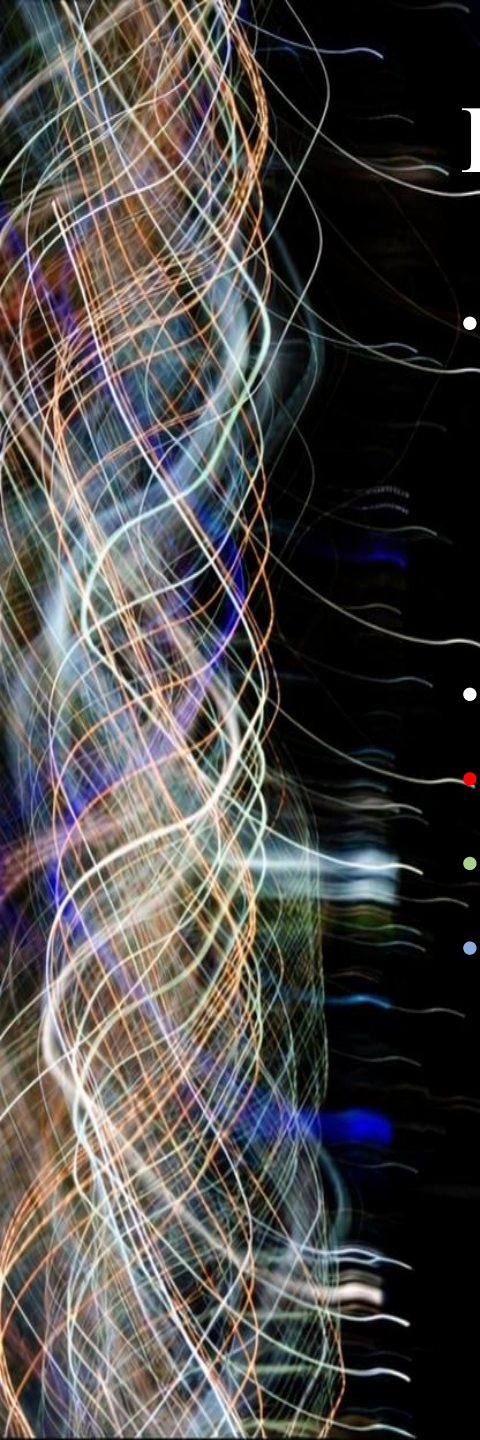
王培儒

2026.04.20



Key Information

- 超導量子計算(Superconducting Quantum Computation, SC QC)，是在極低溫下，以超導(Superconductor)製作的約瑟夫森節(Josephson Junction)為基礎，設計出非線性的LC電路作為量子位元 (Qubit)，搭配電路設計做出可控制、可讀出、可計算的量子處理器(QPU)。整機運作結合微波工程、低溫物理、材料、電路設計與量子控制。
- 04/20：超導量子計算是什麼？
- 04/27：超導量子位元的特性、超導量子電路
- 05/04：如何控制、校正超導量子位元？
- 05/11：超導量子電腦硬體元件導論



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量子模擬



Simulating Physics with Computers

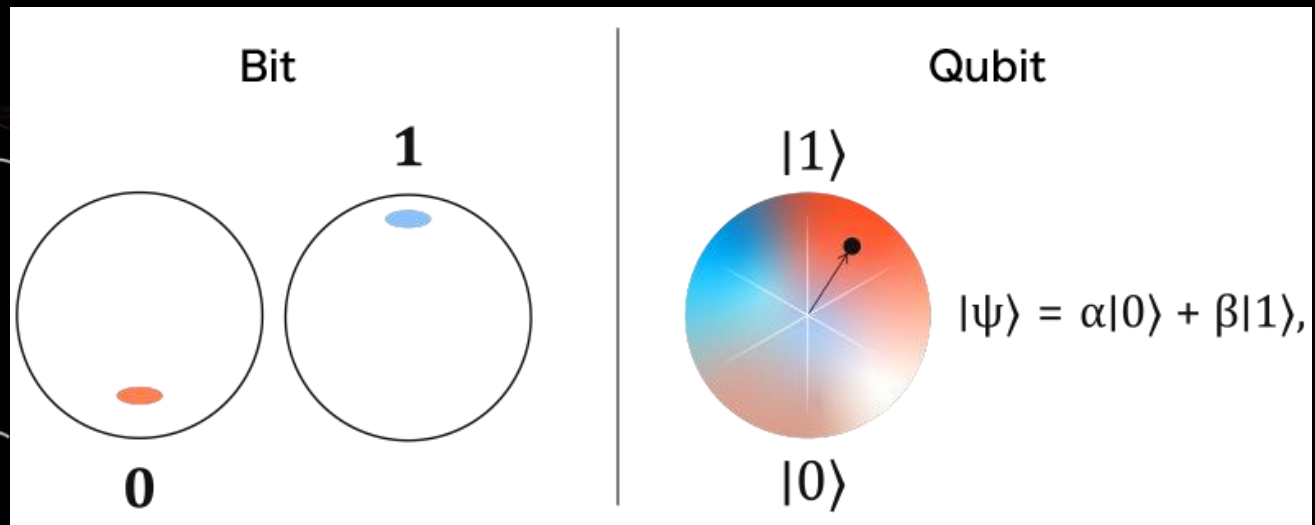
Richard P. Feynman

Department of Physics, California Institute of Technology, Pasadena, California 91107

Received May 7, 1981

be understood very well in analyzing the situation. And I'm not happy with all the analyses that go with just the classical theory, because nature isn't classical, dammit, and if you want to make a simulation of nature, you'd better make it quantum mechanical, and by golly it's a wonderful problem, because it doesn't look so easy. Thank you.

古典位元(Bit)與量子位元(Qubit)



- N個量子位元： 2^N 不同的State
 $|00 \dots 00\rangle, \dots, |01 \dots 10\rangle, \dots, |11 \dots 11\rangle$

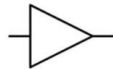
- 疊加態

$$|\psi\rangle = \alpha_1|00 \dots 00\rangle + \dots + \alpha_{2^N}|11 \dots 11\rangle = \sum \alpha_k |k\rangle$$

- 50個Qubit: $2^{50} \times 16 \text{ bytes} \sim 18000 \text{ TB}$

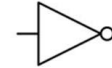
古典邏輯閘(Logic Gate)

Buffer



Input	Output
0	0
1	1

Inverter



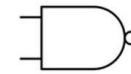
Input	Output
0	1
1	0

AND



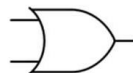
A	B	Output
0	0	0
1	0	0
0	1	0
1	1	1

NAND



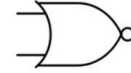
A	B	Output
0	0	1
1	0	1
0	1	1
1	1	0

OR



A	B	Output
0	0	0
1	0	1
0	1	1
1	1	1

NOR



A	B	Output
0	0	1
1	0	0
0	1	0
1	1	0

XOR

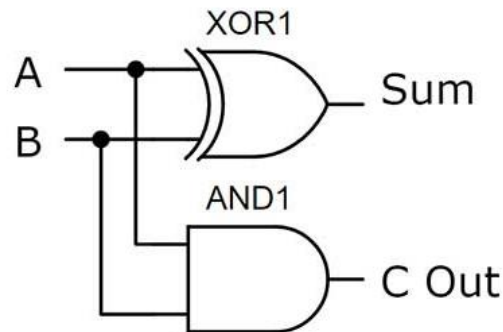


A	B	Output
0	0	0
1	0	1
0	1	1
1	1	0

XNOR



A	B	Output
0	0	1
1	0	0
0	1	0
1	1	1

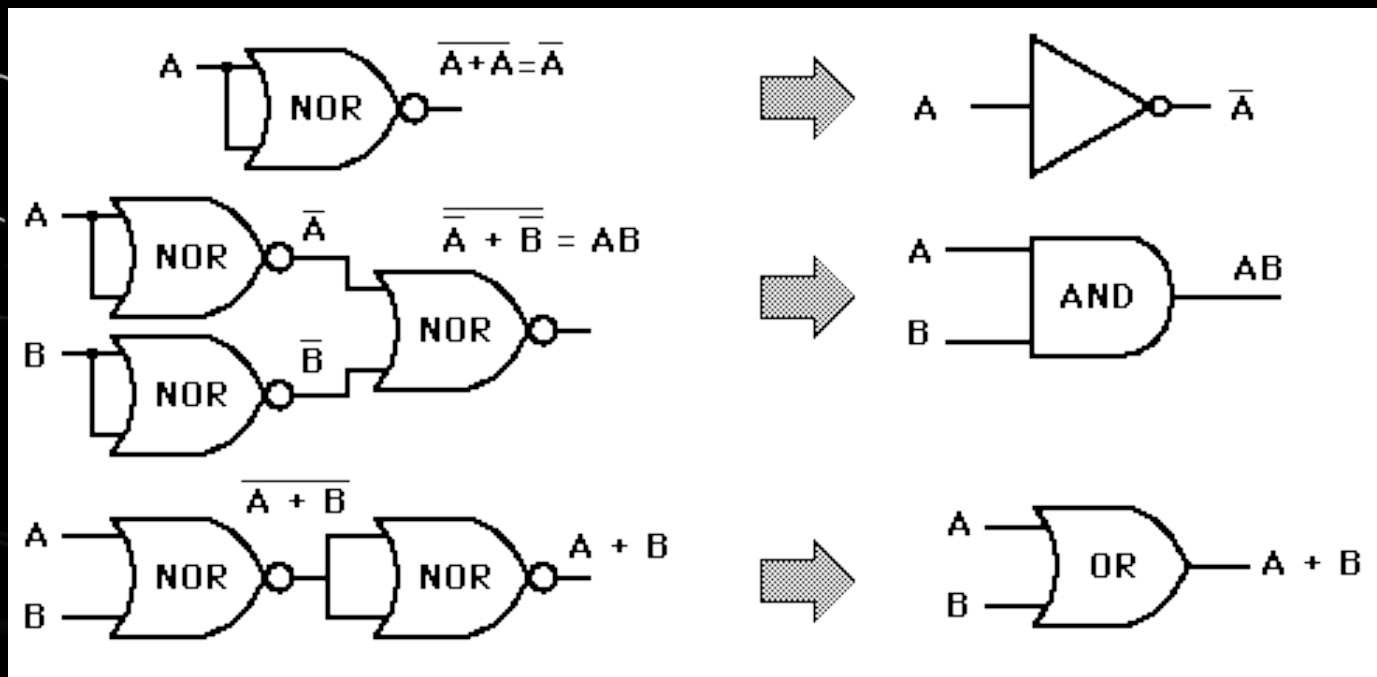


Half Adder

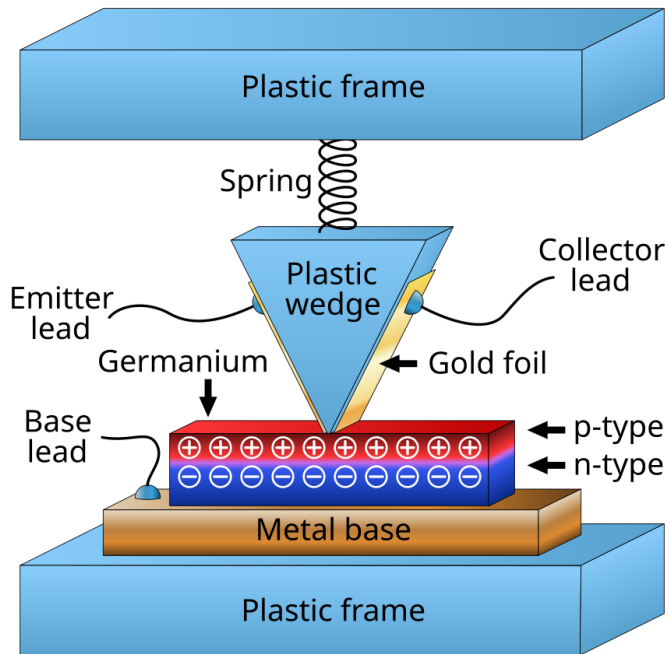
A	B	C Out	Sum
0	0	0	0
1	0	0	1
0	1	0	1
1	1	1	0

通用邏輯閘集(Universal Gate Set)

- 完成所有邏輯運算所需要的「基本積木集合」
- {AND, NOT} 、 {OR, NOT}
- {NAND} 、 {NOR}



電晶體1947



The Nobel Prize in Physics 1956



William Bradford Shockley
Prize share: 1/3



John Bardeen
Prize share: 1/3



Walter Houser Brattain
Prize share: 1/3

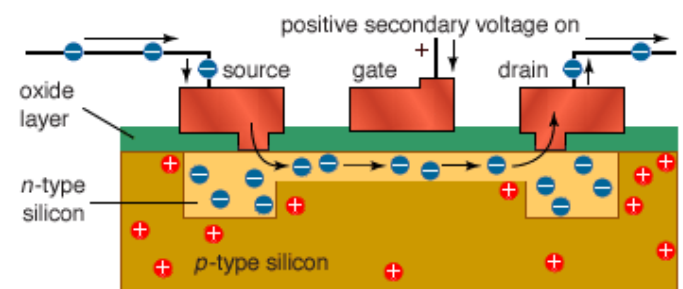
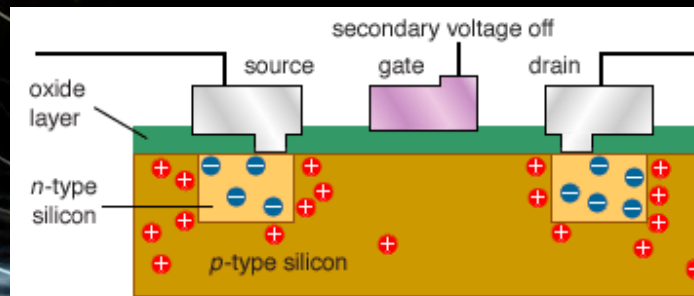
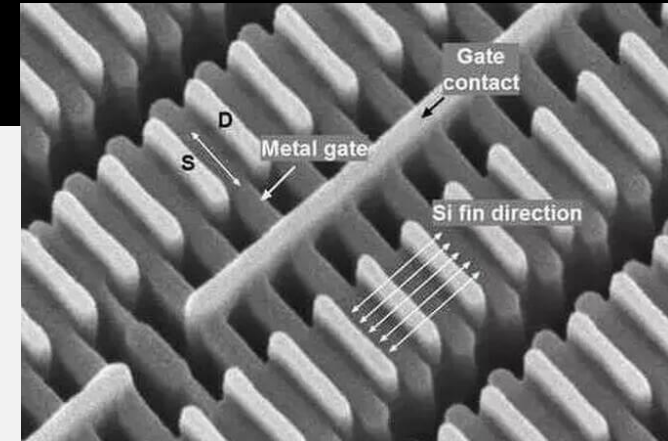
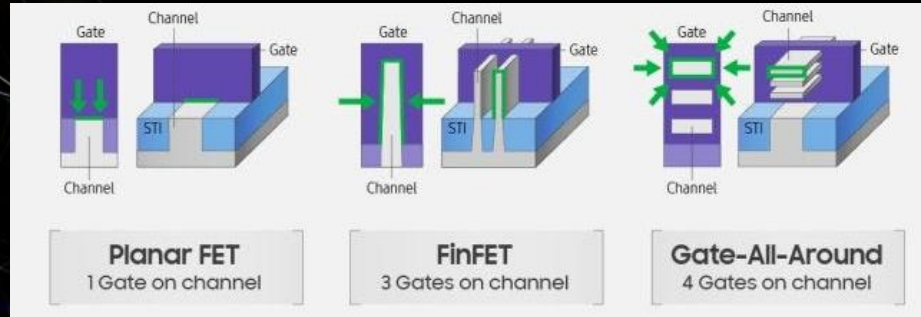
半導體工藝

<https://www.britannica.com/technology/transistor/Junction-transistors#ref228688>

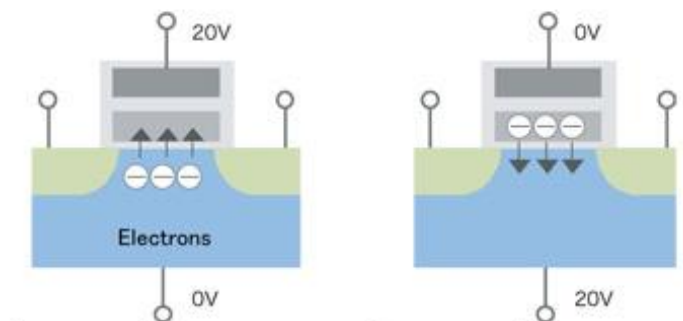
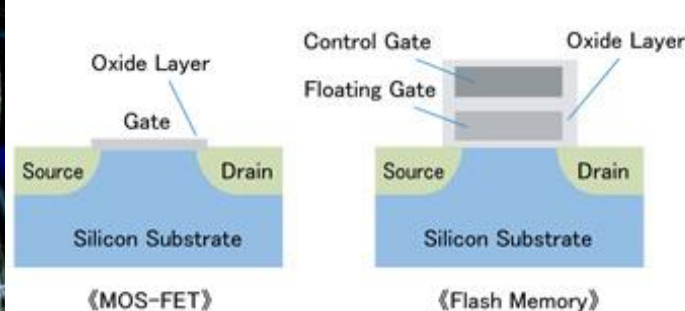
<https://wiki.csie.ncku.edu.tw/embedded/Flash>

<https://www.utmel.com/blog/categories/transistors/introduction-to-finfet>

<https://deepforest.substack.com/p/a-crash-course-in-transistors>



MOS-FET and flash memory construction



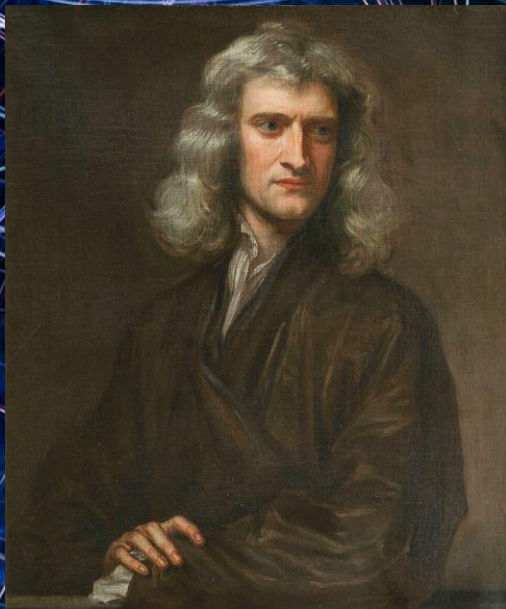
《Write operation》
A voltage applied to the control gate causes a tunnel current to flow through the oxide layer, thereby injecting electrons into the floating gate.

《Erase operation》
A voltage applied to the silicon substrate releases the electrons accumulated at the floating gate.



可逆計算與量子閘

物理定律(通常)是可逆的



Newton

$$F = \dot{p}$$



Hamilton

$$\dot{q} = \frac{\partial H}{\partial p}$$
$$\dot{p} = -\frac{\partial H}{\partial q}$$



Schrodinger

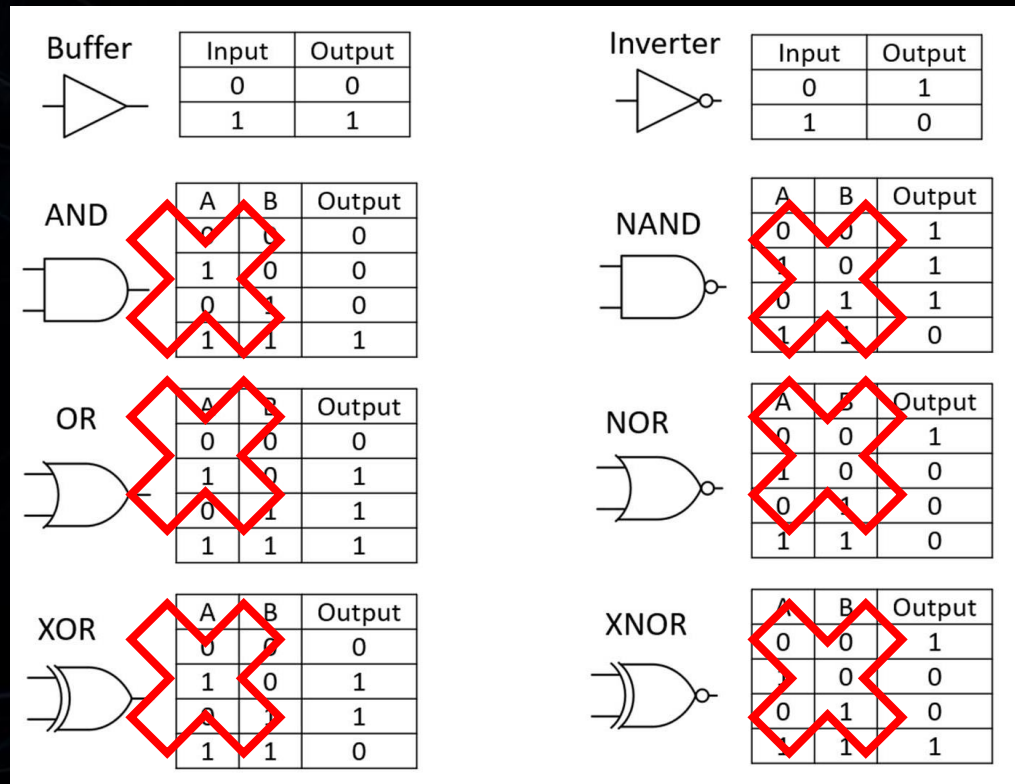
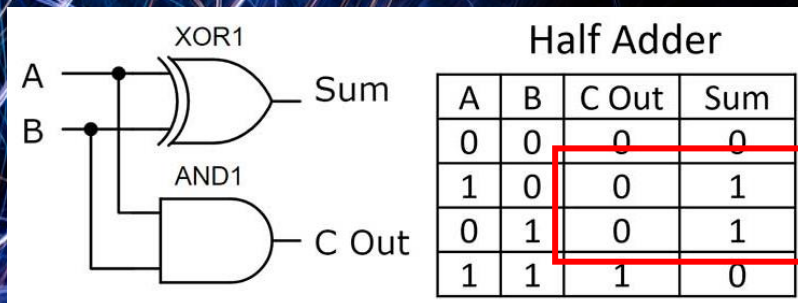
$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi$$

$$\psi(t) = \left[\exp \left(-\frac{i}{\hbar} T \left[\int_0^t \hat{H}(s) ds \right] \right) \right] \psi(0)$$

資訊抹除與不可逆性

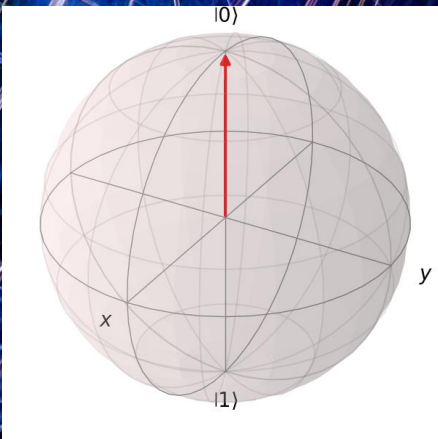
- 抹除或喪失1 bit的資訊，會至少產生 $k_B \ln 2$ 的熵
(或需要耗能 $E \geq k_B T \ln 2$)

Landauer

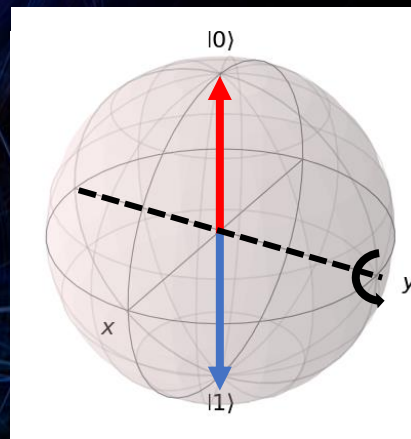


可逆邏輯閘-Single Qubit Gate

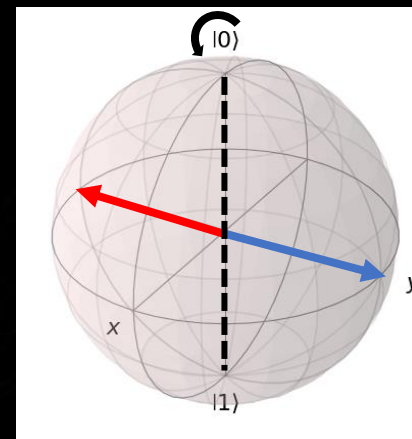
X Gate (NOT Gate)			$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$ 0\rangle \rightarrow 1\rangle, 1\rangle \rightarrow 0\rangle$
Y Gate			$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$	
Z Gate			$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	
Hadamard Gate		Superposition	$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$	$ 0\rangle \rightarrow \frac{1}{\sqrt{2}}(0\rangle + 1\rangle)$



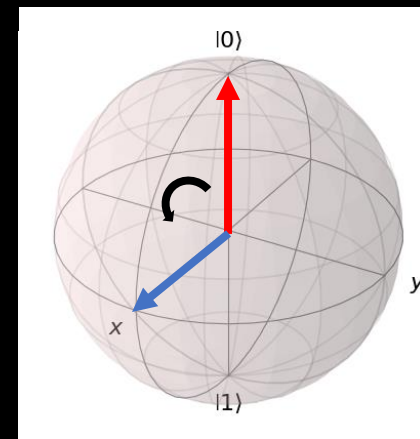
X Gate (NOT Gate)



Y



Z

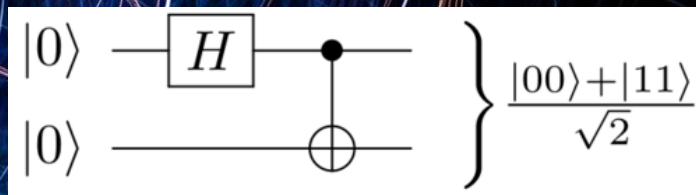
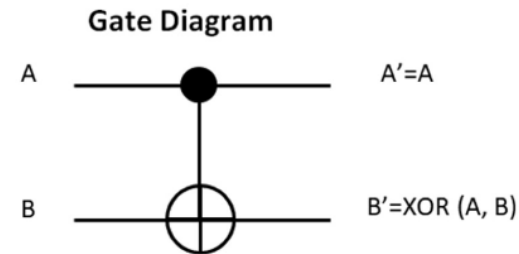


H

可逆邏輯閘-Multi Qubit Gate

Control-NOT Gate (CNOT)/Control-X Gate (CX)

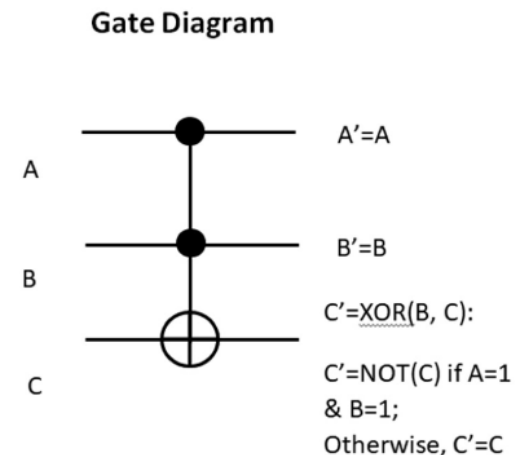
Logic Table			
A	B	A'	B'
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0



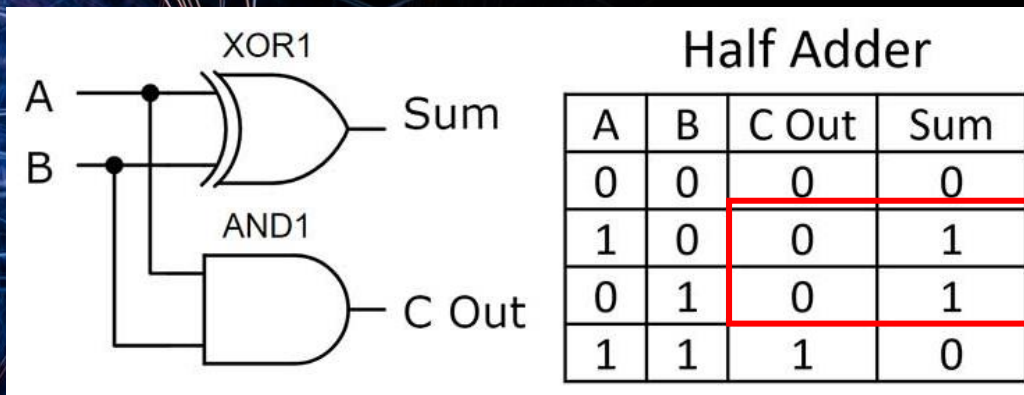
$$|0\rangle_A |0\rangle_B \rightarrow \frac{(|0\rangle_A + |1\rangle_B)}{\sqrt{2}} |0\rangle_B \rightarrow \frac{(|0\rangle_A |0\rangle_B + |1\rangle_B |1\rangle_B)}{\sqrt{2}}$$

Control-Control-NOT Gate (CCNOT)/Toffoli Gate

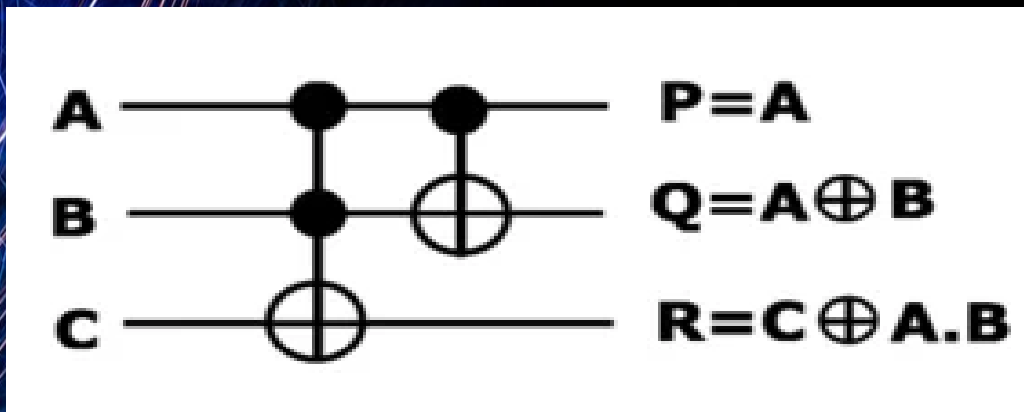
Logic Table					
A	B	C	A'	B'	C'
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	0
0	1	1	0	1	1
1	0	0	1	0	0
1	0	1	1	0	1
1	1	0	1	1	1
1	1	1	1	1	0



可逆加法器

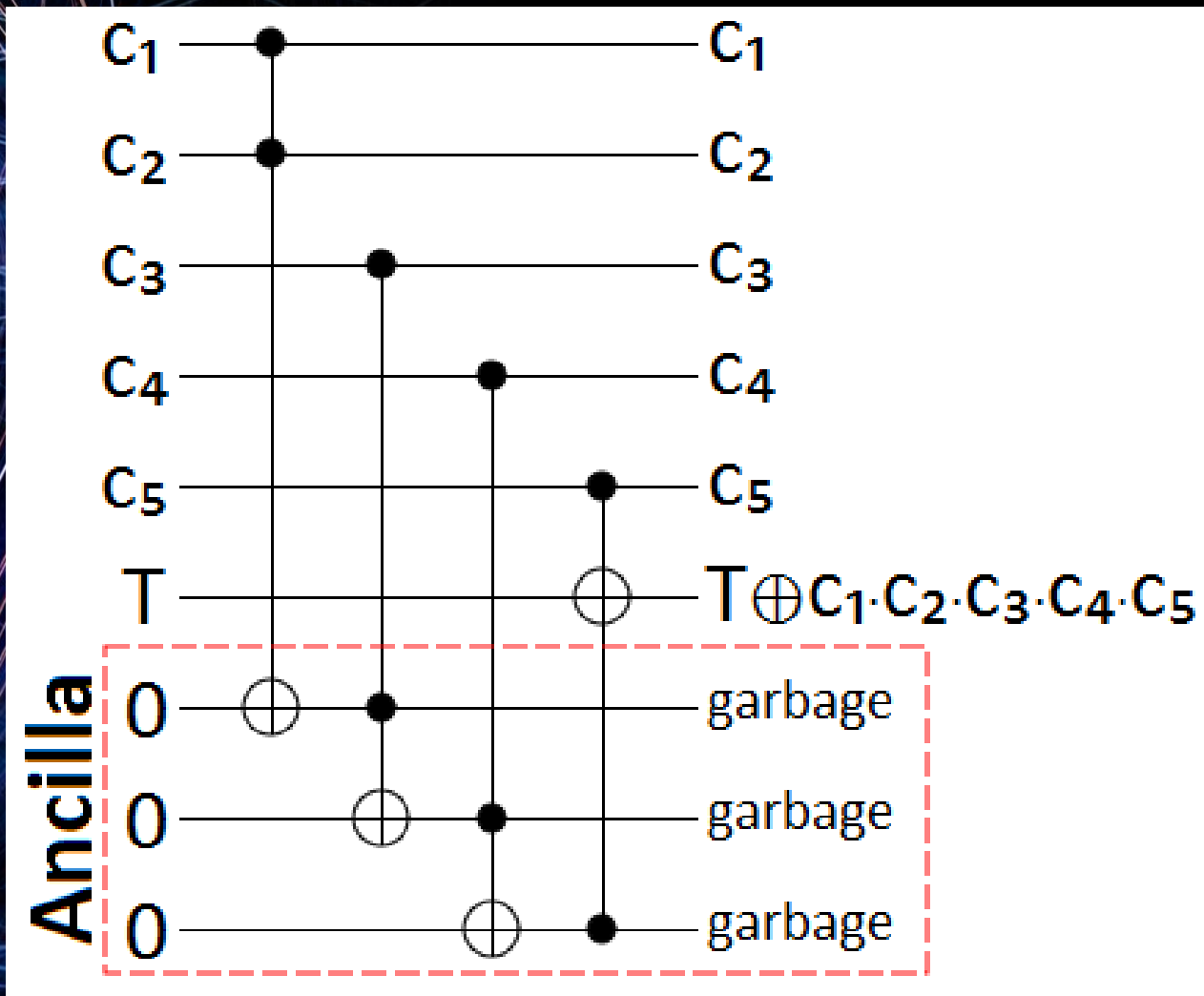


	B	A	C'	S
	0	0	0	0
	1	0	0	1
	0	1	0	1
	1	1	1	0



C	B	A	C'	B'	A'
0	0	0	0	0	0
0	1	0	0	1	0
0	0	1	0	1	1
0	1	1	1	0	1

Ancilla Bit



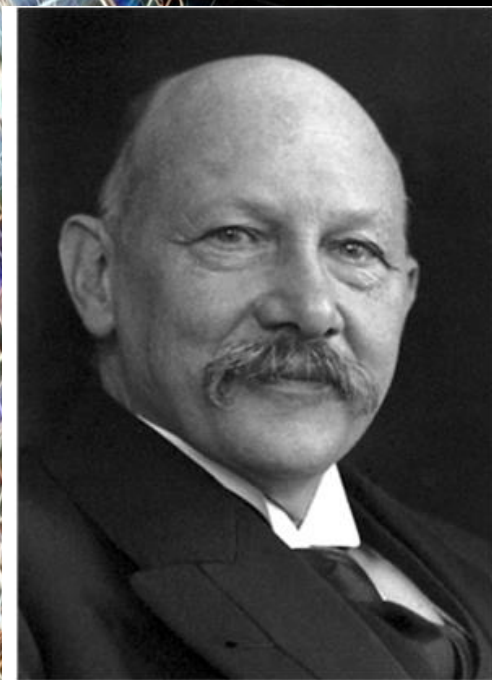
Short Summary

- 微觀物理定律是可逆的
- 計算過程如果不可逆，會有損耗
- 需要將傳統邏輯電路改成可逆邏輯電路
- **Ancilla**：為了保留資訊，需要額外的輔助位元

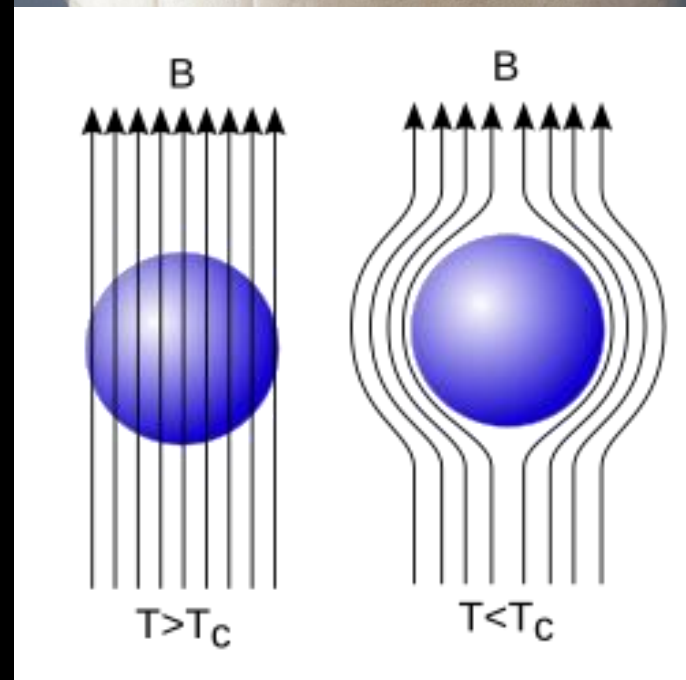
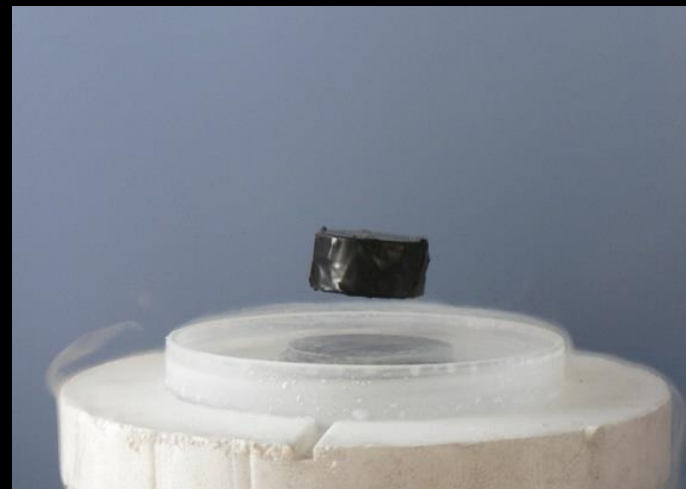
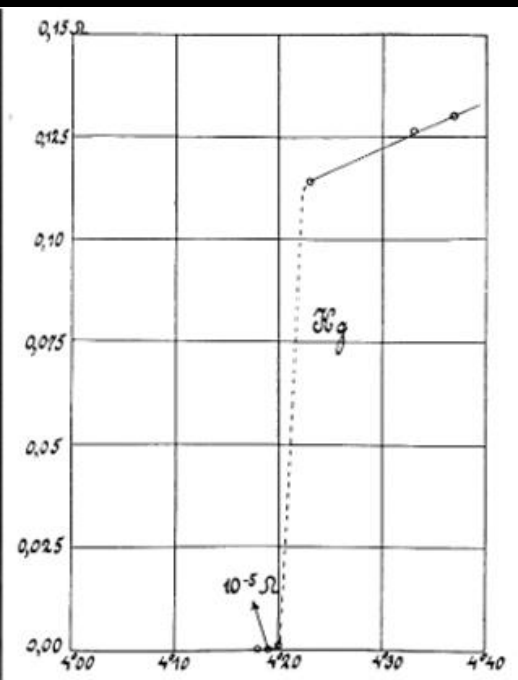


超導量子位元

超導體 Superconductor



Kamerlingh Onnes
Hg的超導性, 1911
1913 Nobel



Meissner Effect
完全抗磁性, 1933

純元素超導性

alkali
metals

PERIODIC TABLE OF SUPERCONDUCTING ELEMENTS

18

1 H	2 He																	18					
3 Li	4 Be																	13 B	14 C	15 N	16 O	17 F	18 Ne
11 Na	12 Mg																	13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr						
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe						
55 Cs	56 Ba	57 La	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn						
87 Fr	88 Ra	89 Ac																103 Lr					
			58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu							
			90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr							

alkaline earth metals

transition metals

metals

semi-metals

non-metals

Atomic number

Symbol

Critical temperature of bulk at normal pressure

Critical temperature under certain conditions

Condition type (e. g. pressure value, film form)

superconducting element only under certain conditions (pressure or film form)

superconducting element at normal pressure in bulk form

Be: 0.026 K, 9.95 K film

C: 15 K nanotube

O: 0.6 K, 120 GPa

S: 17 K, 160 GPa

Se: 7 K, 13 GPa

Br: 1.4 K, 150 GPa

I: 1.2 K, 25 GPa

Xe: 1.2 K, 25 GPa

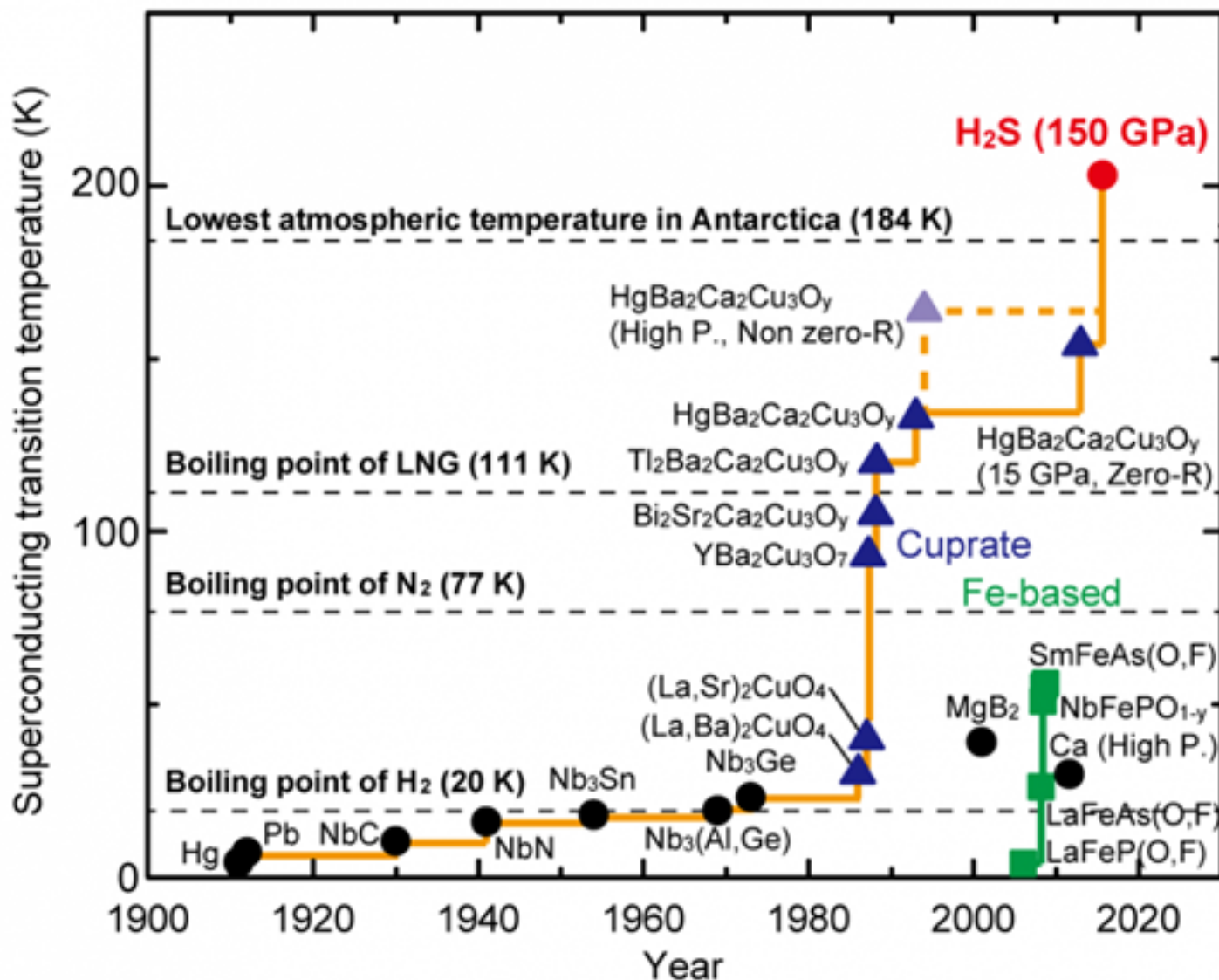
At: 1.2 K, 25 GPa

Rn: 1.2 K, 25 GPa

Lu: 0.1 K, 1.2 K, 18 GPa

Am: 1 K

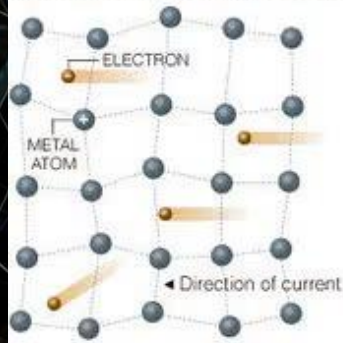
超導材料與臨界溫度



BCS理論

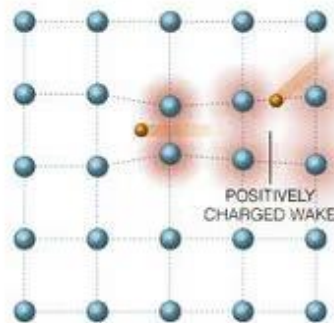
Low-Temperature Superconductivity

December was the 50th anniversary of the theory of superconductivity, the flow of electricity without resistance that can occur in some metals and ceramics.



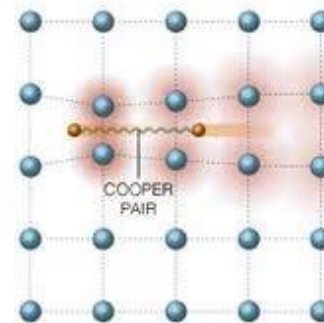
ELECTRICAL RESISTANCE

Electrons carrying an electrical current through a metal wire typically encounter resistance, which is caused by collisions and scattering as the particles move through the vibrating lattice of metal atoms.



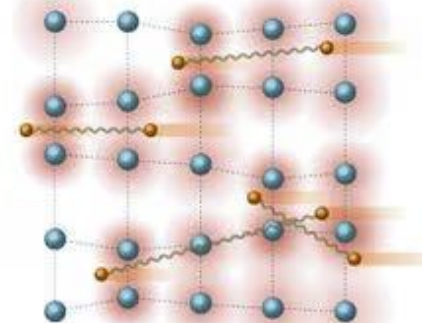
CRITICAL TEMPERATURE

As the metal is cooled to low temperatures, the lattice vibration slows. A moving electron attracts nearby metal atoms, which create a positively charged wake behind the electron. This wake can attract another nearby electron.



COOPER PAIRS

The two electrons form a weak bond, called a Cooper pair, which encounters less resistance than two electrons moving separately. When more Cooper pairs form, they behave in the same way.



SUPERCONDUCTIVITY

If a pair is scattered by an impurity, it will quickly get back in step with other pairs. This allows the electrons to flow undisturbed through the lattice of metal atoms. With no resistance, the current may persist for years.

1975 Nobel



Bardeen, Cooper, and Schrieffer, 1957

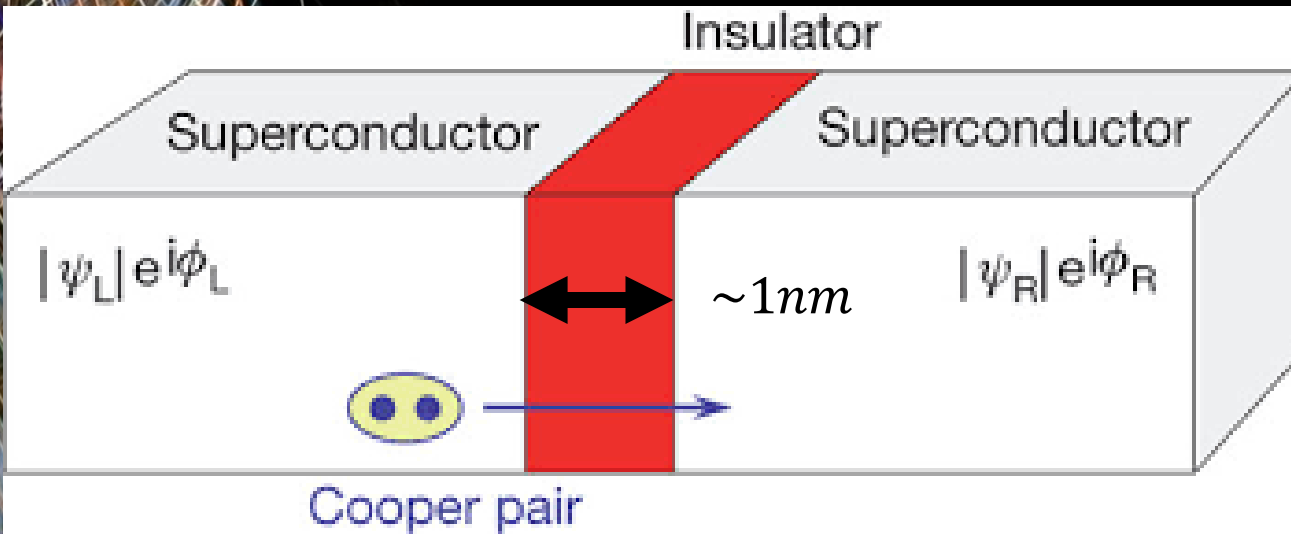
$$\Psi = |\Psi|e^{i\theta}$$

約瑟夫森節 Josephson Junction

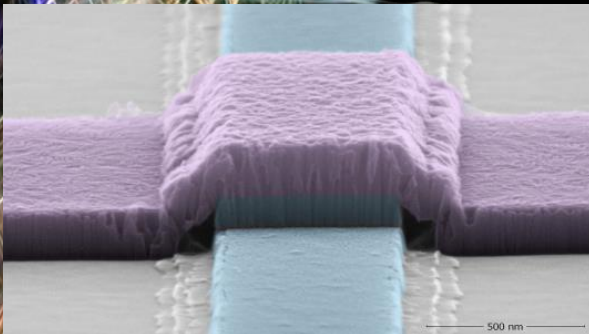
$$\Delta\phi = \phi_L - \phi_R$$

$$I(\Delta\phi) = I_c \sin \Delta\phi$$

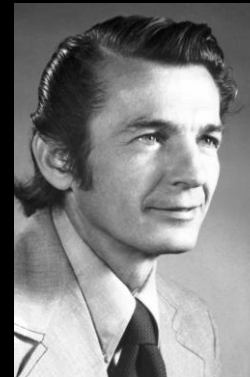
$$\begin{aligned} V &= \frac{h}{2e} \frac{1}{I_c \cos \Delta\phi} \frac{dI}{dt} \\ &= \frac{h}{2e} \frac{1}{\sqrt{I_c^2 - I^2}} \frac{dI}{dt} \\ &\equiv L(I) \frac{dI}{dt} \end{aligned}$$



1973 Nobel



Josephson
超導相位穿隧理論
的開創性工作(1962)



Giaever
超導中穿隧與證實
BCS超導能隙(1960)

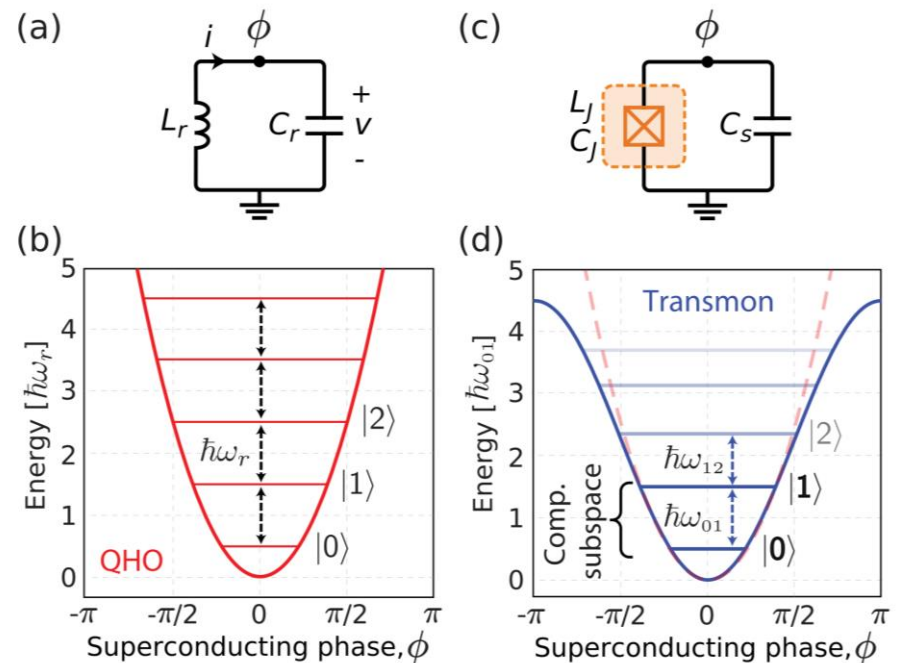


Leo Esaki
江崎 玲於奈
半導體穿隧(1957)

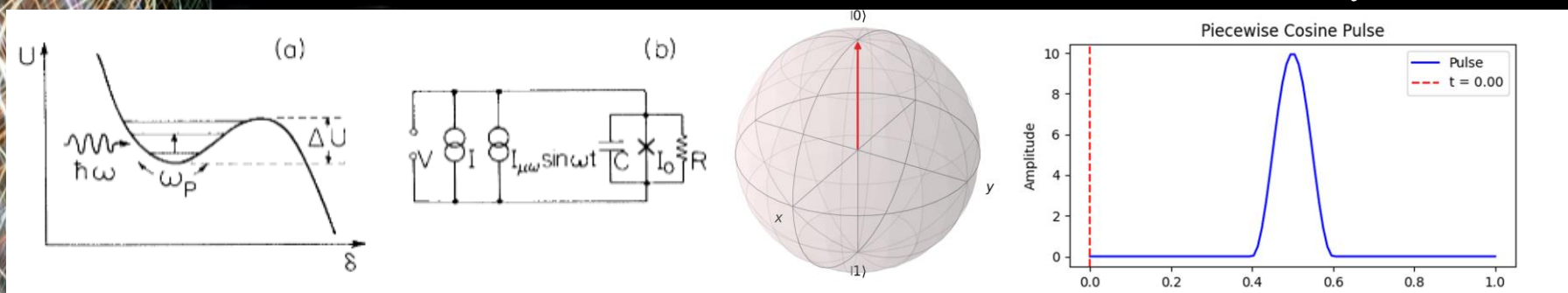
超導量子位元-非線性LC電路



1985



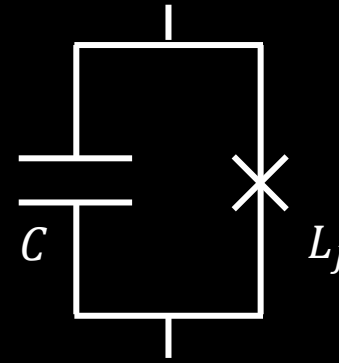
Anharmonicity



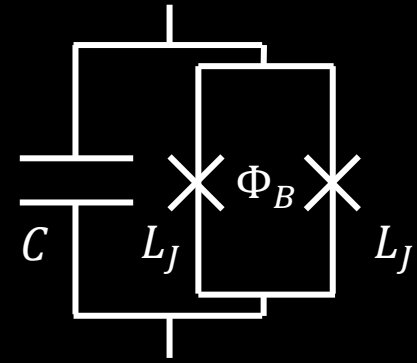
Tunable Transmon (SQUID)

Superconducting QUantum
Interference Device

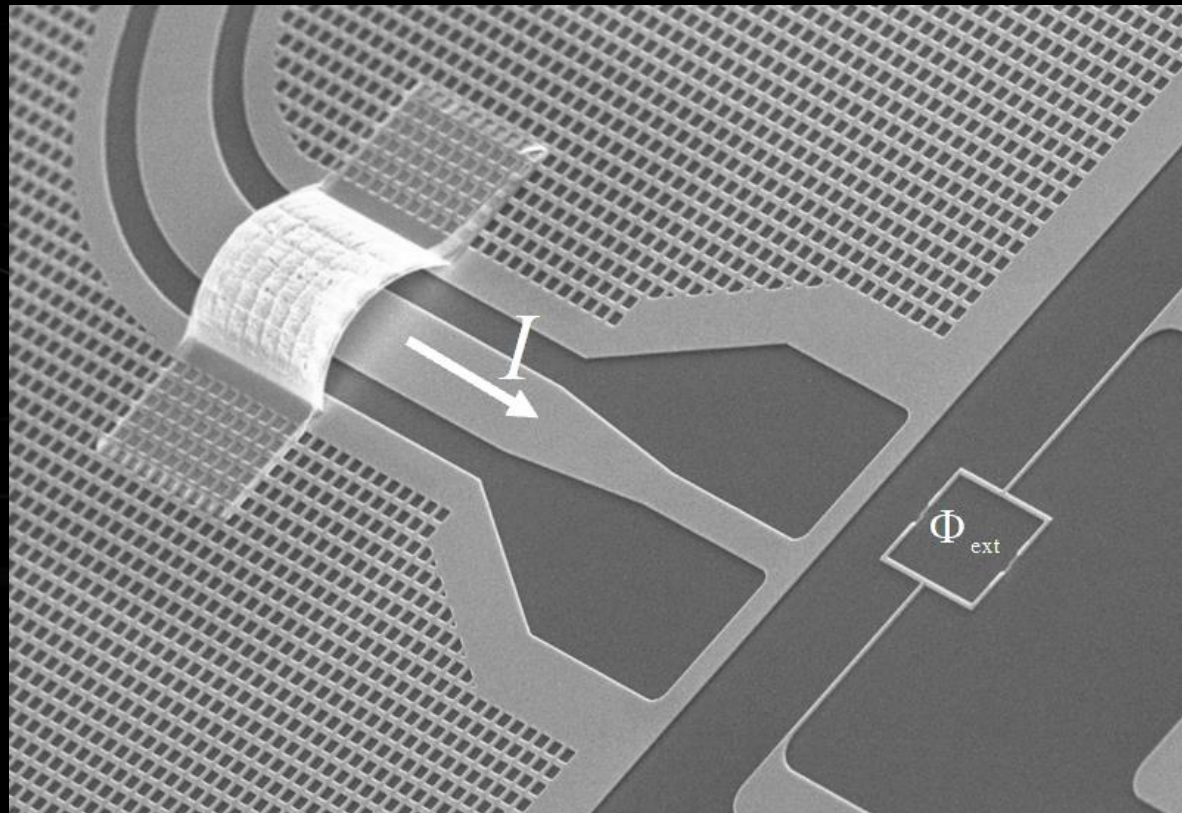
$$L \rightarrow L(\Phi_B)$$



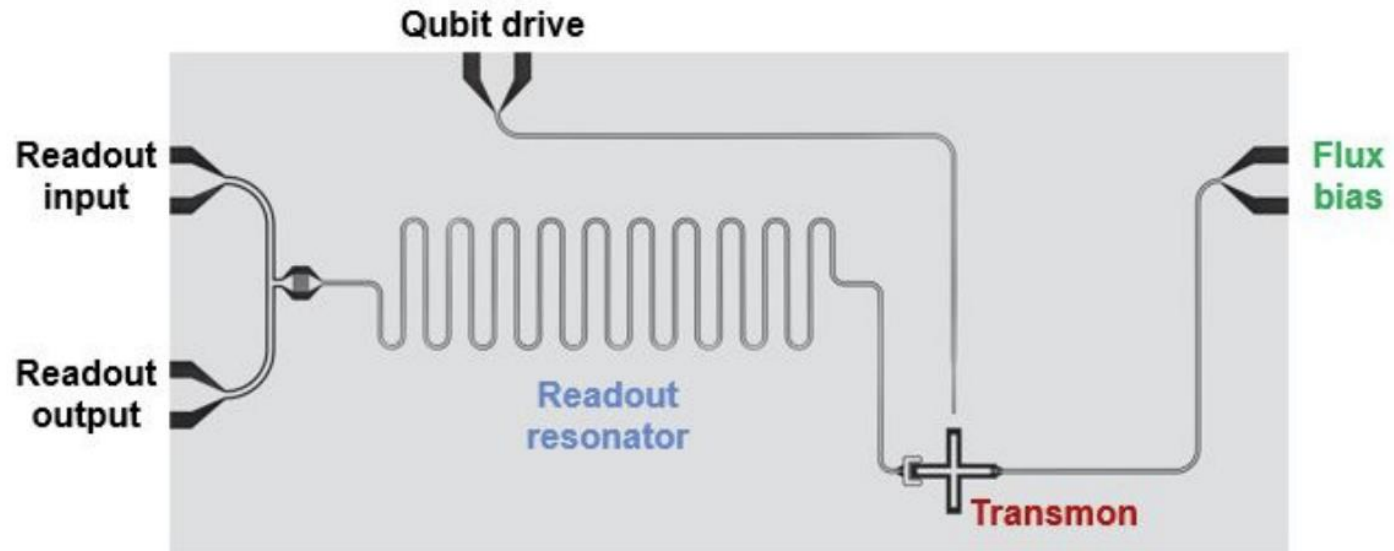
Transmon



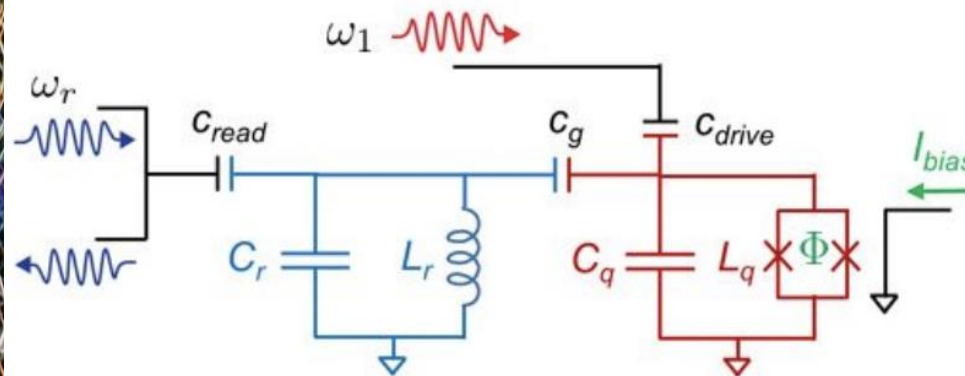
Tunable Transmon



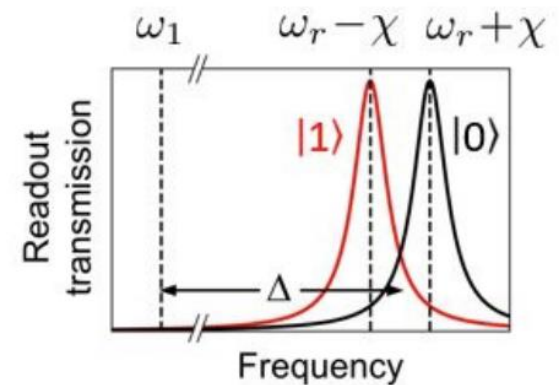
Single Qubit Circuit



(a)

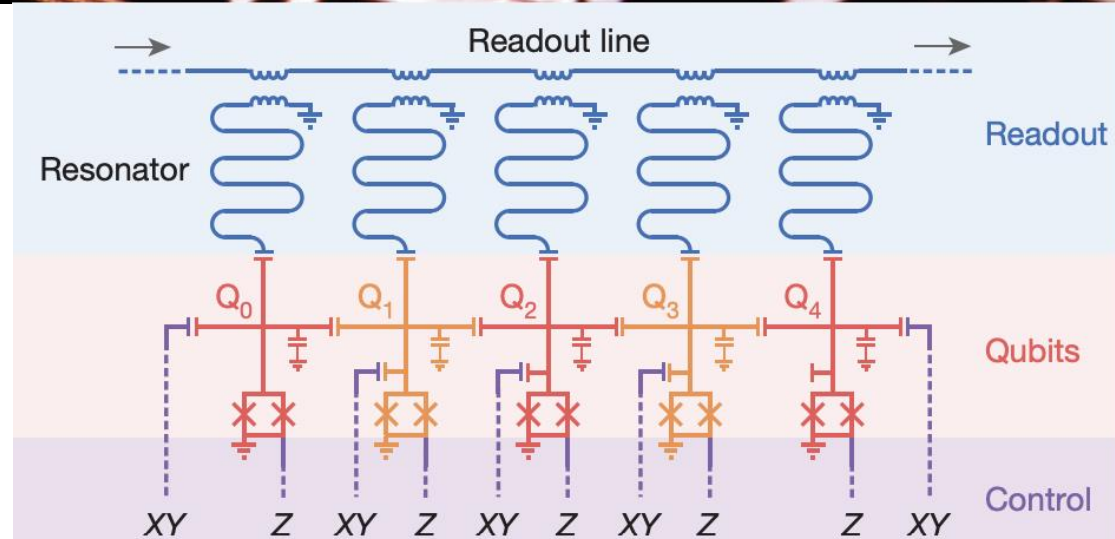
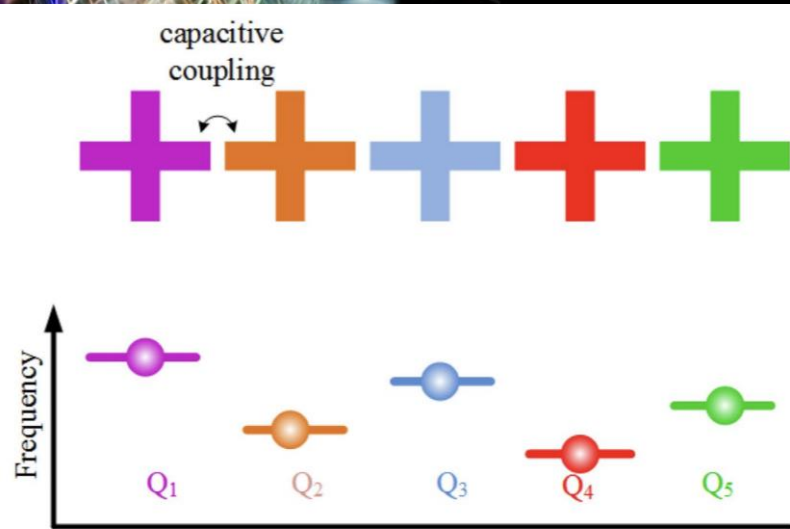
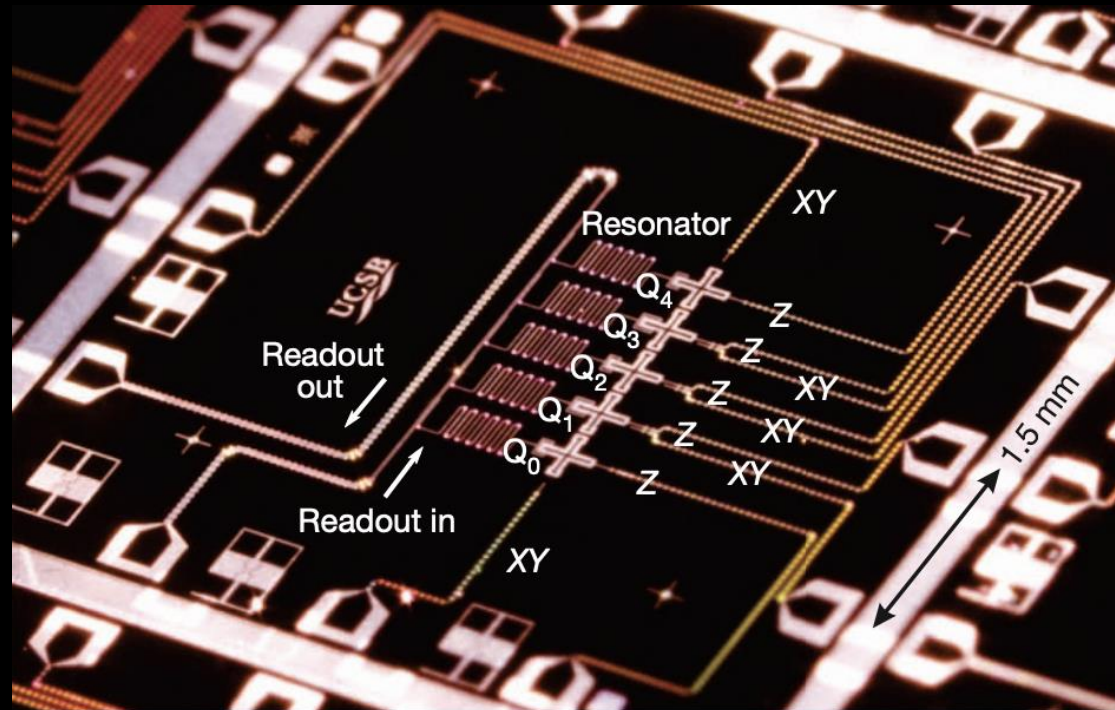


(b)



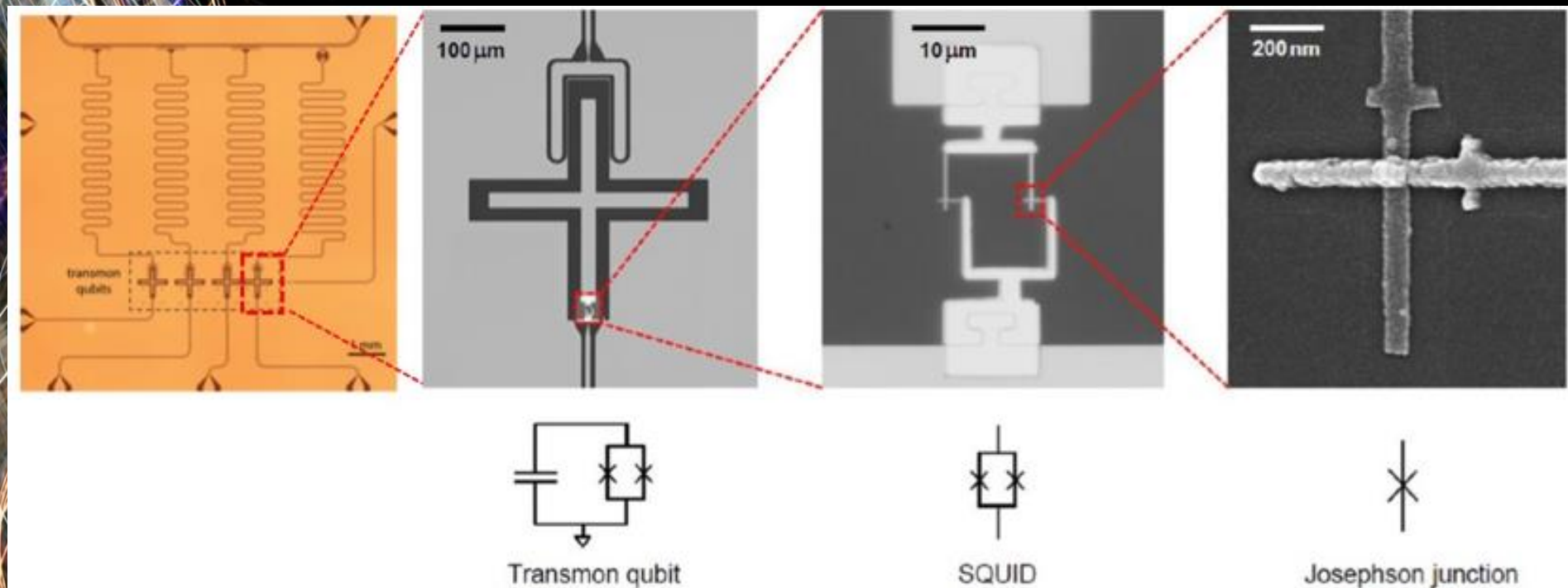
(c)

Multi Qubit Circuit



Short Summary

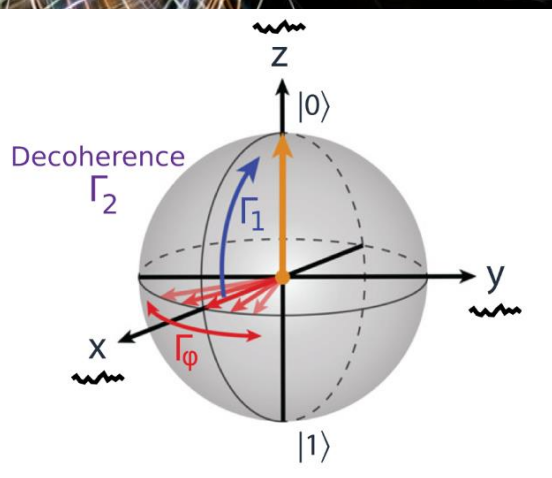
- 一個超導量子位元可由一個電容+約瑟芬森節組成的非線性LC電路。可調式量子位元由雙約瑟芬森節組成。
- 非線性效應使得能階差異化(Anharmonicity)，可獨立作為計算空間。
- 量子位元搭配聯外的控制線，可進行控制與讀取





噪聲與糾錯

退相干(decoherence)與噪聲(noise)

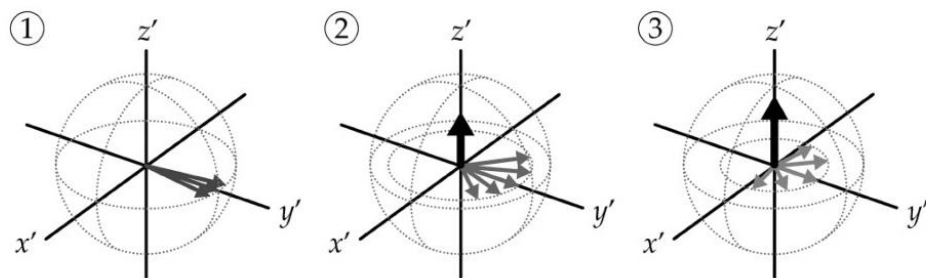


T_1 :relaxation time

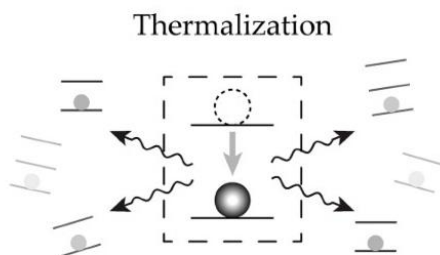
T_2 :dephasing time

$$\rightarrow \Gamma_i = \frac{1}{T_i} \text{“rate”}$$

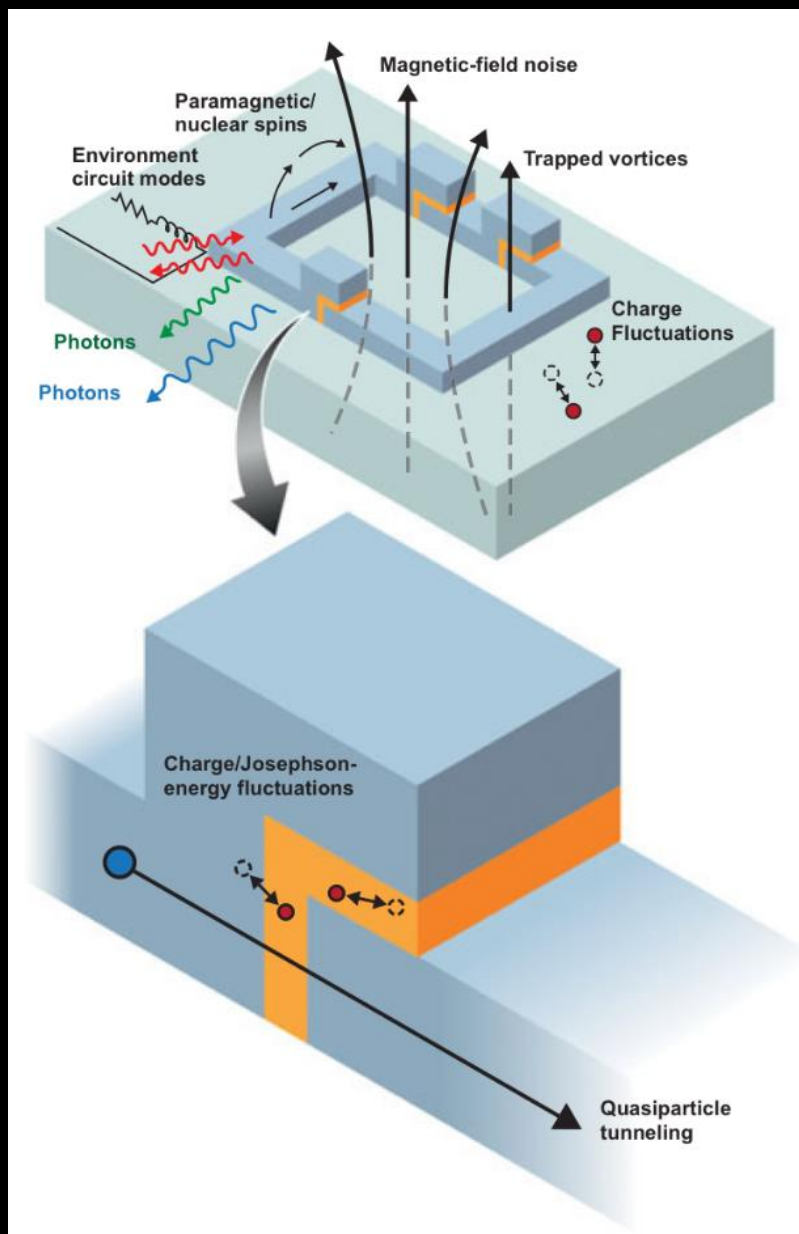
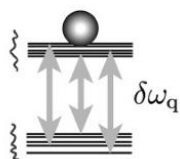
(a) Relaxation in the Bloch sphere



(d) Mechanisms

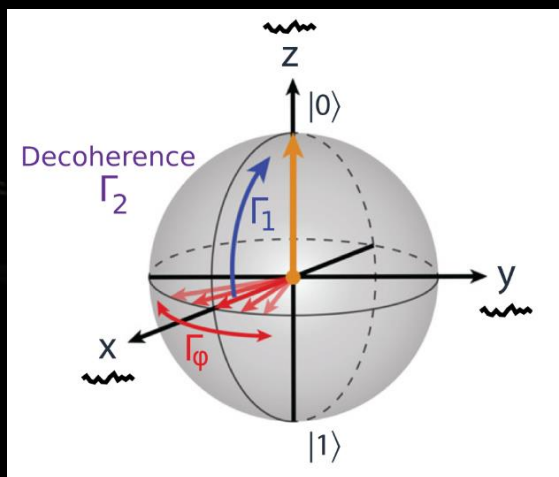


Dephasing

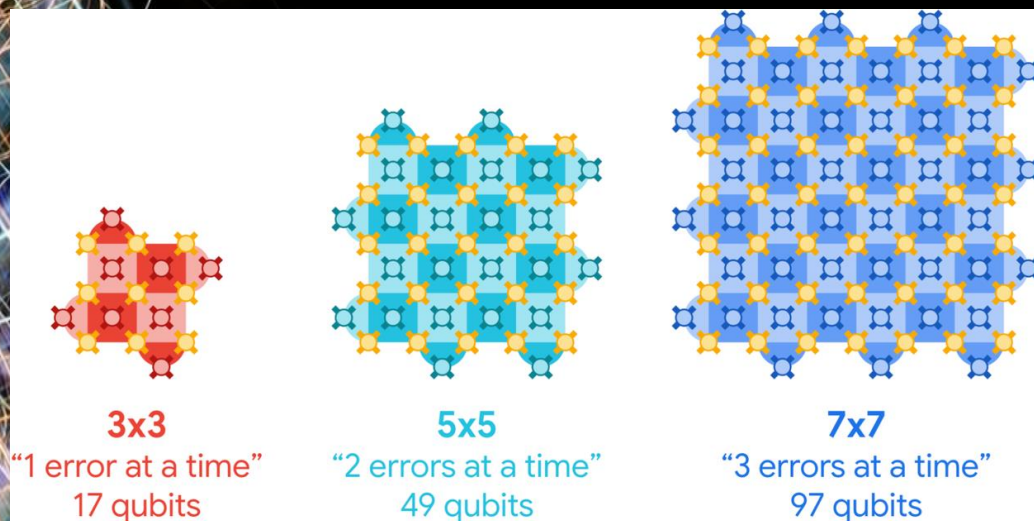


量子糾錯 Quantum Error Correction

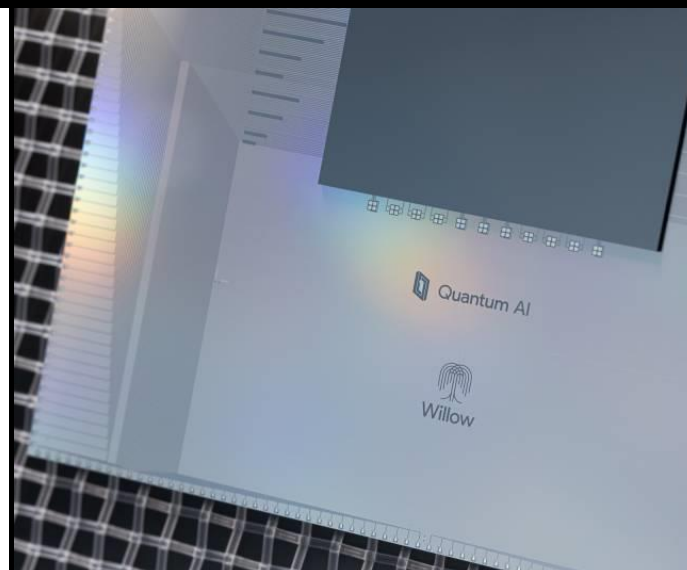
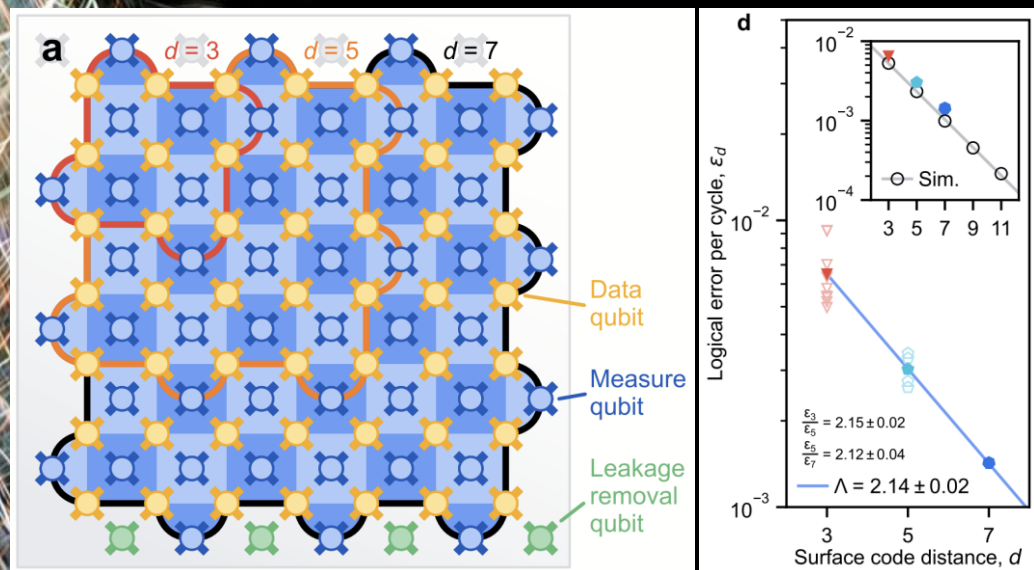
- 古典只有Bit Flip Error : $0 \leftrightarrow 1$
- 透過冗余編碼，將多個物理位元組成一個「邏輯位元」
 $\{0\} \rightarrow \{000\}, \{1\} \rightarrow \{111\}$ 可以容許一個錯誤
 $\{0\} \rightarrow \{00000\}, \{1\} \rightarrow \{11111\}$ 可以容許兩個錯誤
- 古典可以「測量」來檢查是否有出錯
- 古典可以複製備份數據
- 量子不可直接量測、也不可複製備份
- 量子有複雜的錯誤發生



量子糾錯Quantum Error Correction

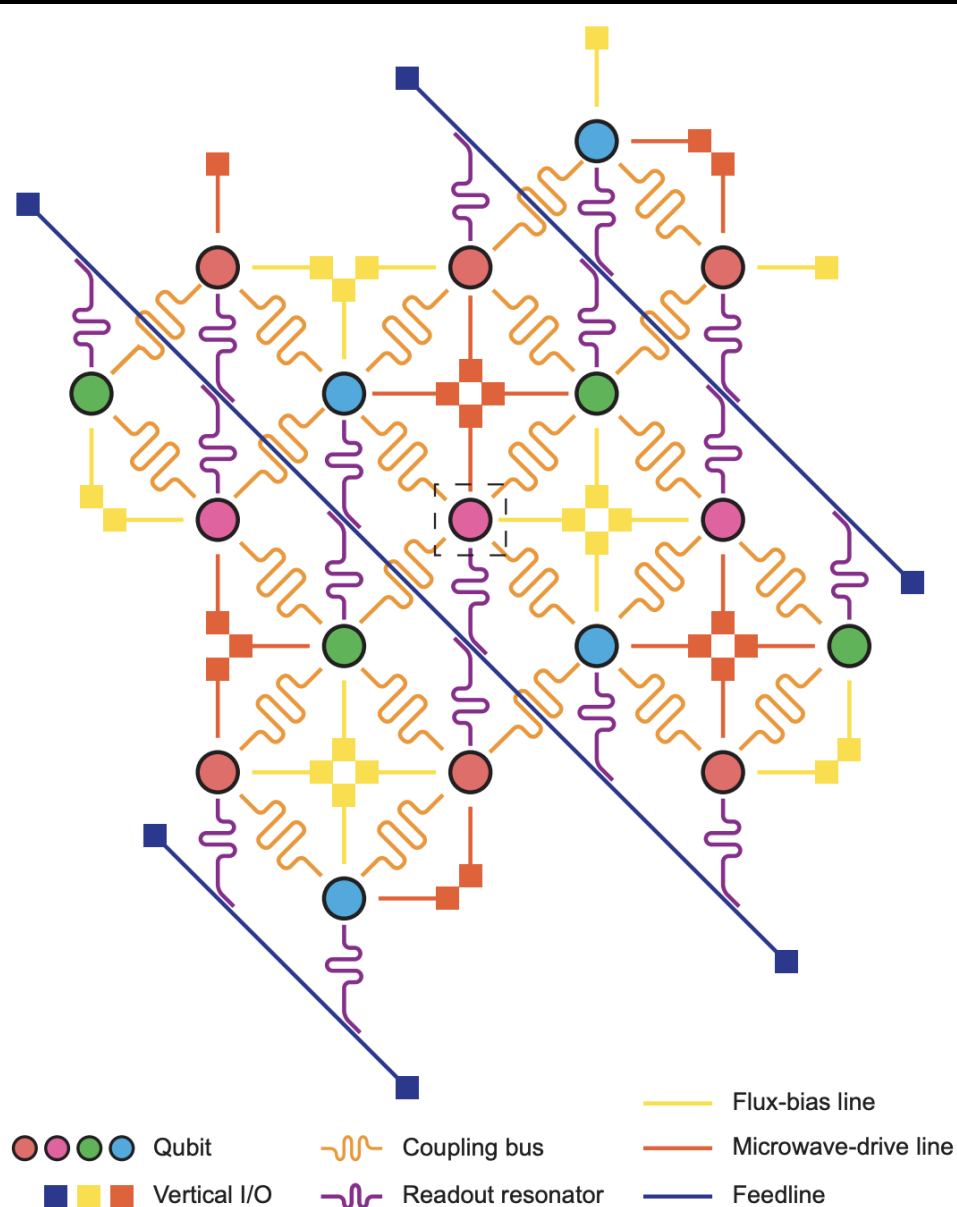
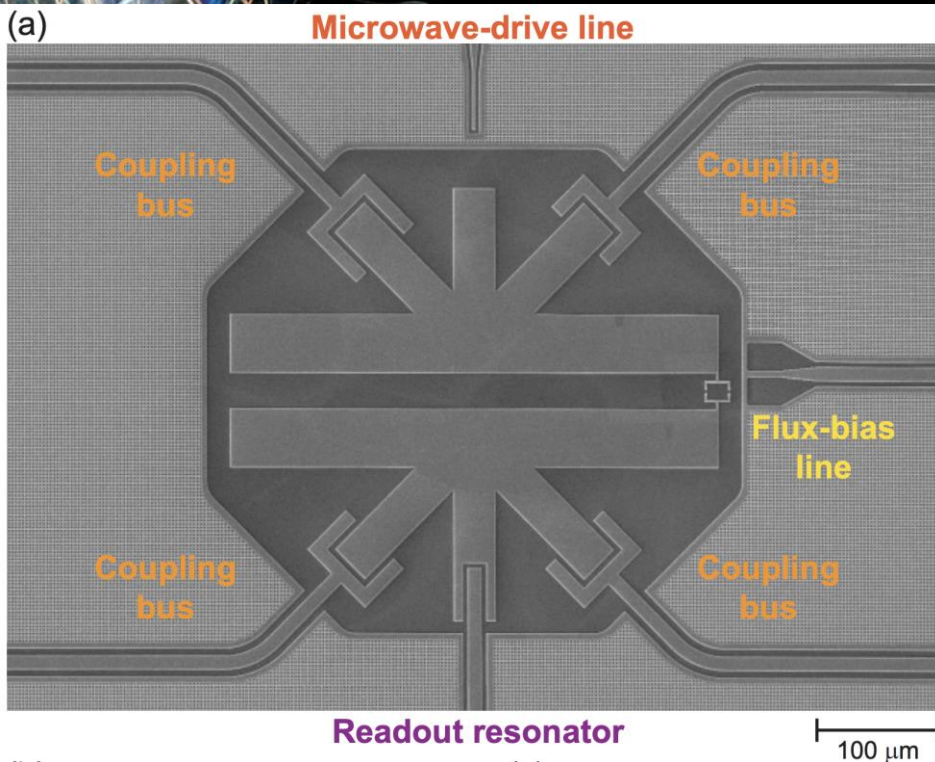


- 表面碼(Surface Code)是一種適合2D量子晶片的QEC
- 透過多個位元組成邏輯位元，並加入Ancilla，透過測量Ancilla來間接檢查錯誤

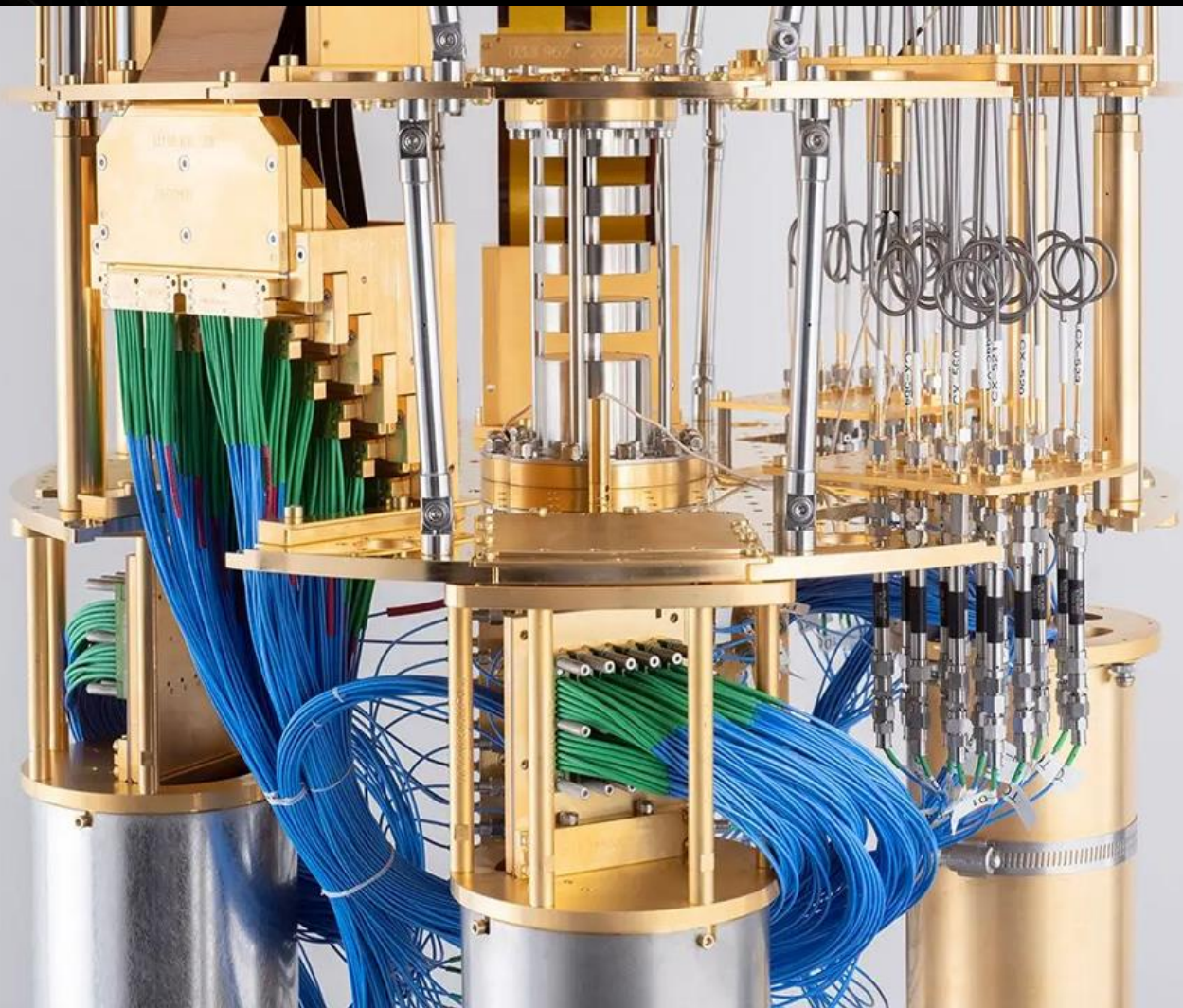


表面碼糾錯晶片

- 7 wires per qubit
- Crosstalk

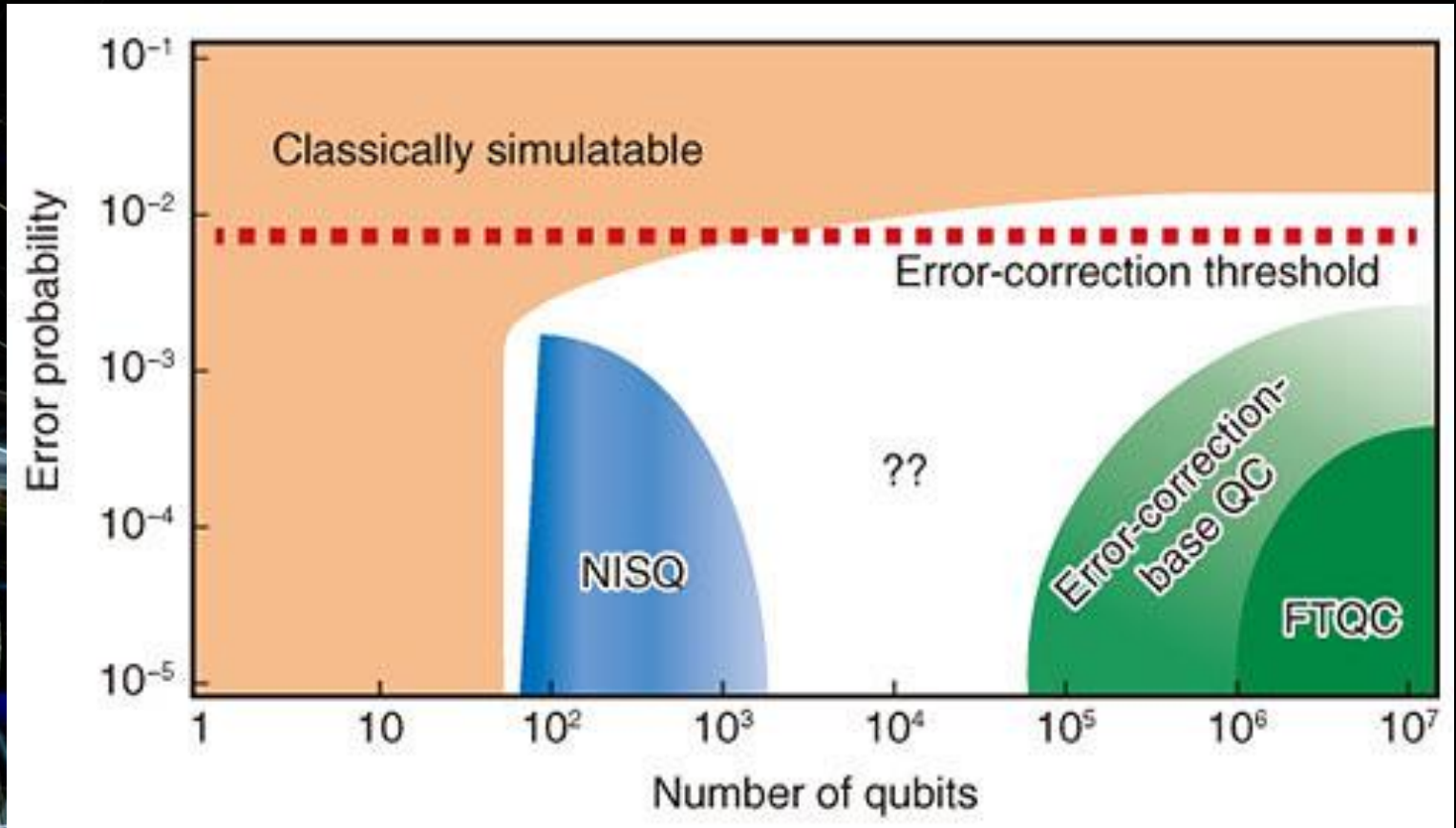


複雜走線

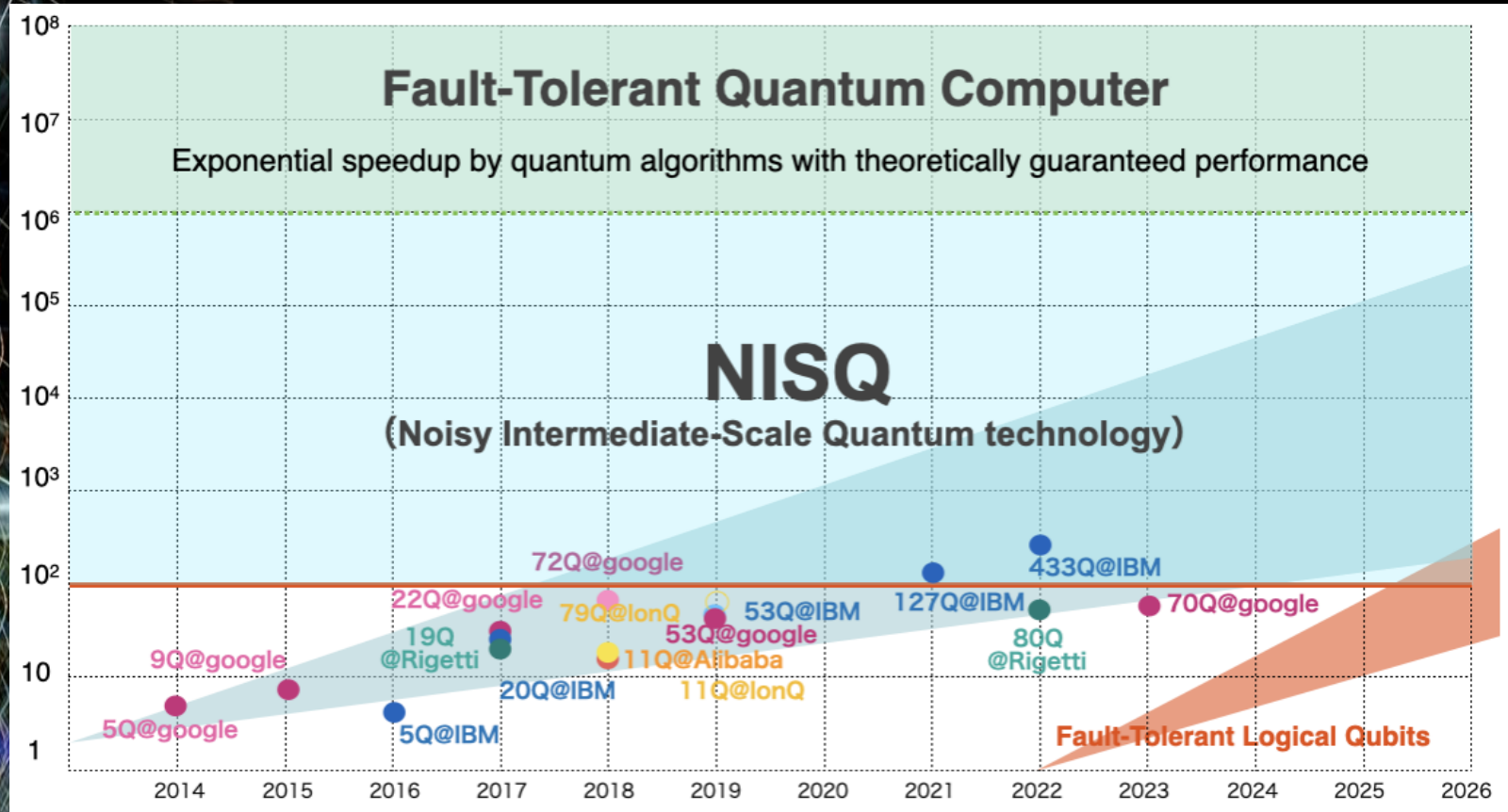


NISQ和FTQC

- NISQ: Noisy Intermediate-Scale Quantum computing
- FTQC: Fault-Tolerant Quantum Computing

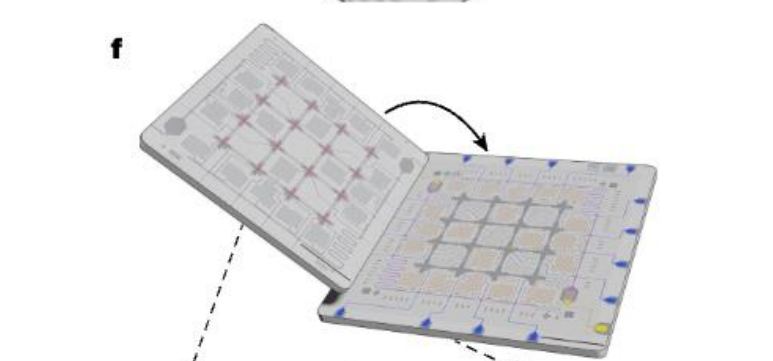
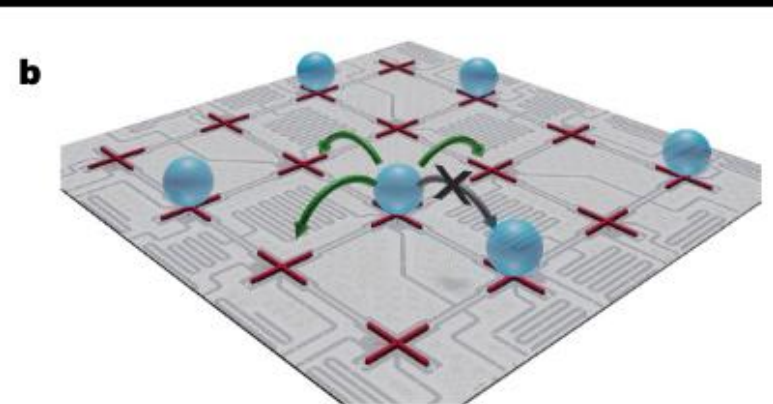
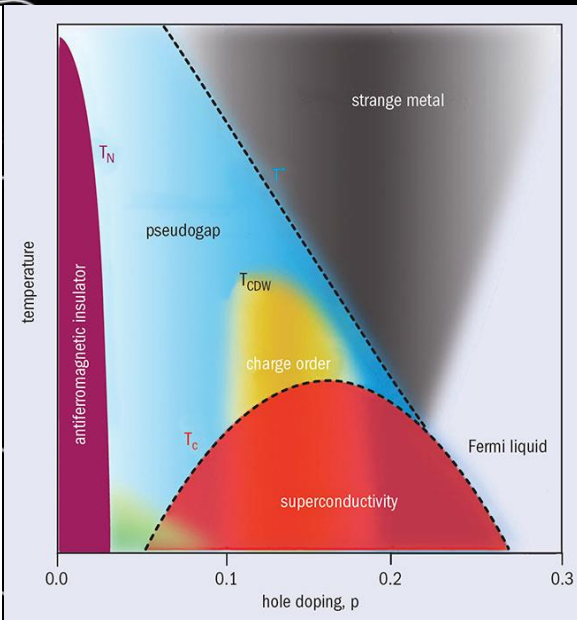
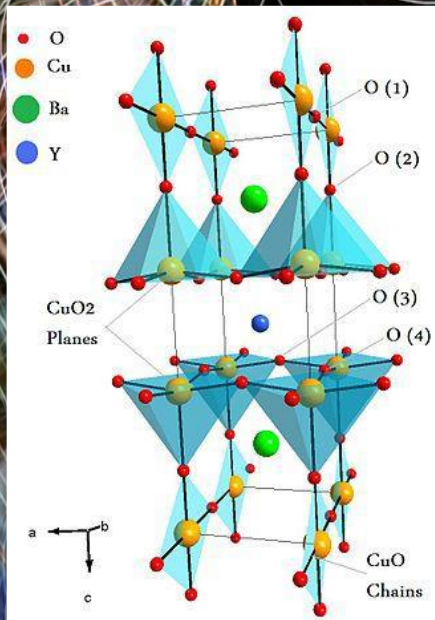


NISQ和FTQC



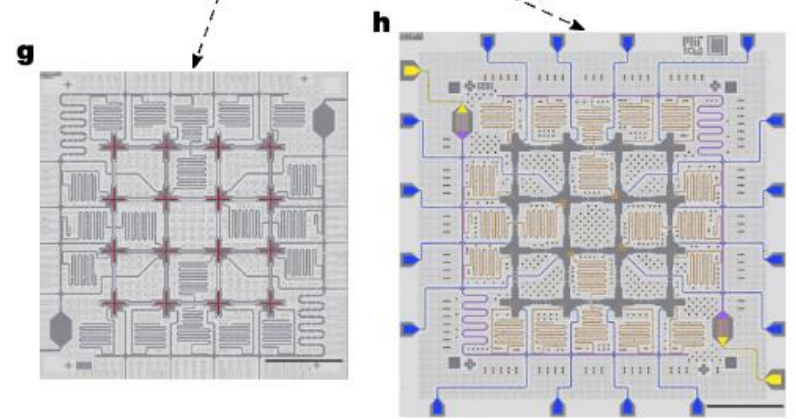
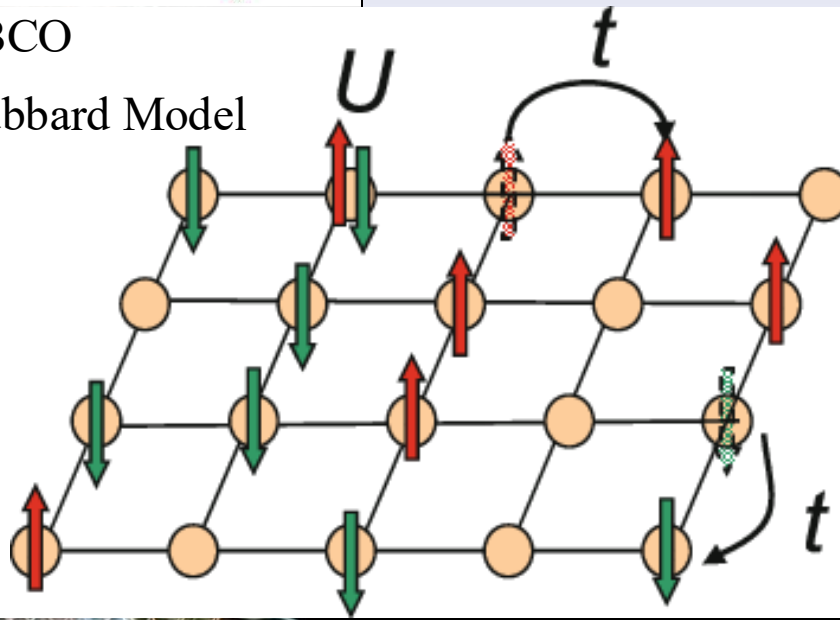
材料模擬

<https://doi.org/10.1038/s41586-024-07325-z>
<https://commons.wikimedia.org/wiki/File:Ybco.jpg>
https://en.wikipedia.org/wiki/Hubbard_model#/media/File:2D-Hubbard-model.png
<https://cemcourier.com/a/taming-high-temperature-superconductivity/>



YBCO

Hubbard Model



Legend: Qubit (red), Resonator (orange), Flux line (blue), Purcell filter (purple), Feedline (yellow).



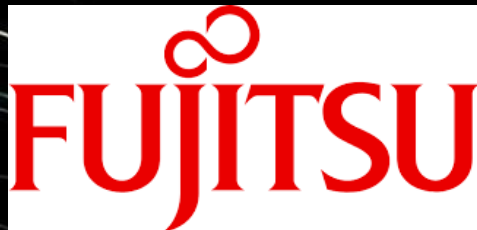
Short Summary

- 量子位元極易受到干擾出錯，需要QEC進行糾錯
- 具備完整QEC能力的量子計算稱為FTQC，需要百萬(>1M)個物理量子位元構成，現今仍不可行
- NISQ可以在少量位元與可接受的錯誤下，執行部分量子近似優化算法

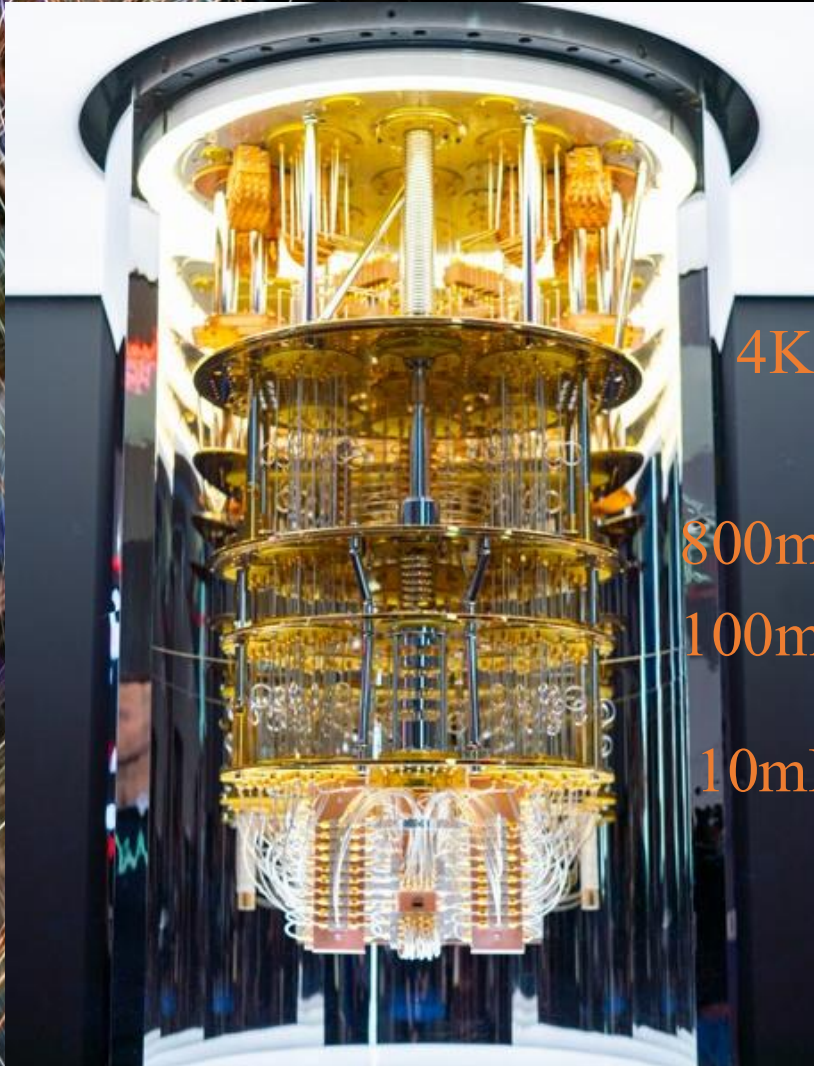


量子計算原型機

Top Players in SC QC



IBM's Prototype

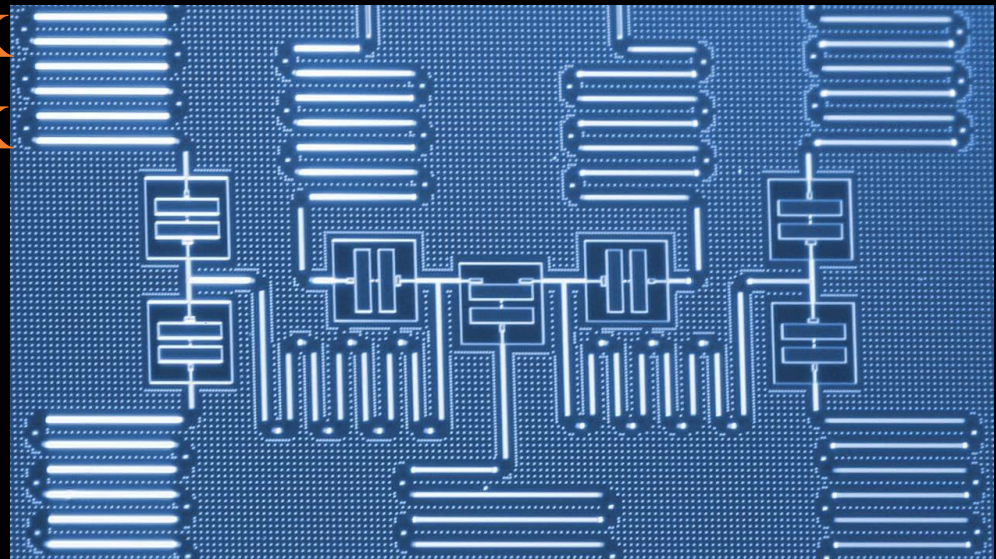


4K

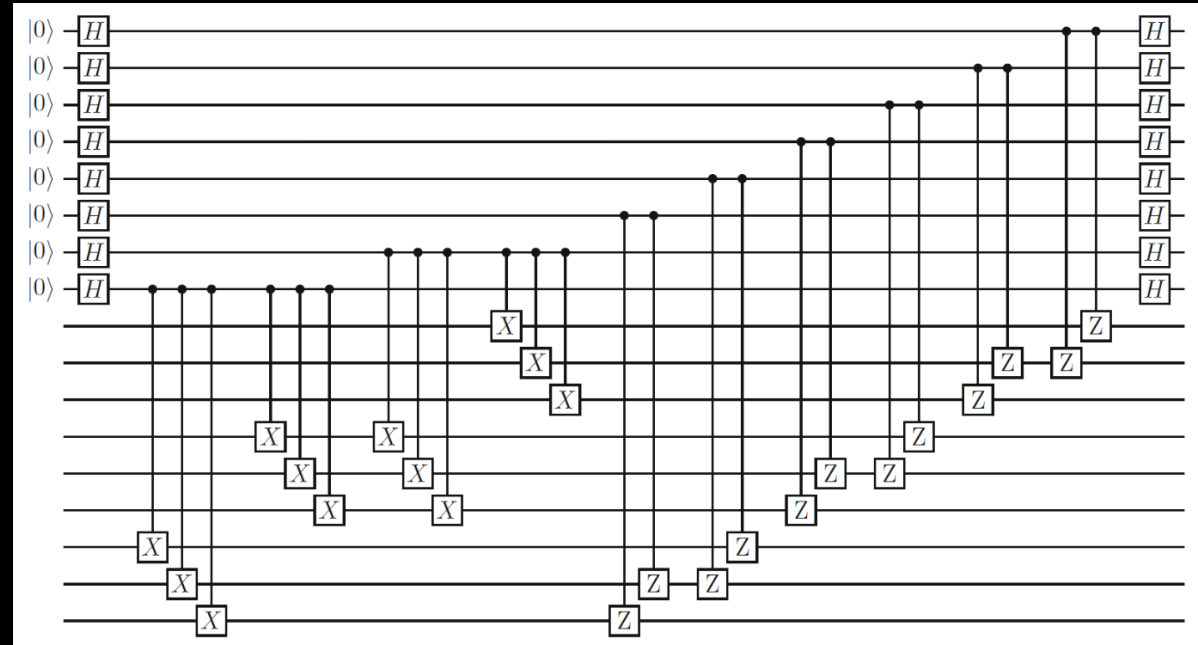
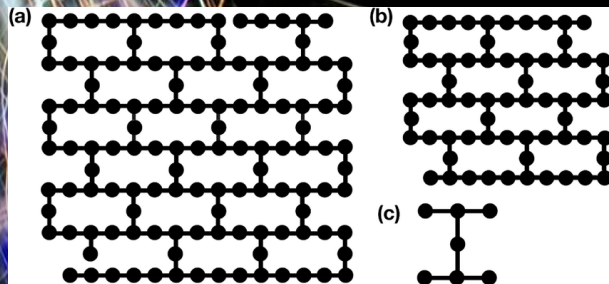
800mK

100mK

10mK



QPU Architecture



Processor	Penguin v1	Penguin v2	Penguin v3	Penguin v4	Falcon r4
IBM					
Avg. qubit connectivity	3.9	3.7	2.3	2.3	2.1

Figure 1 consists of two parts. Part (a) is a schematic diagram of a 2D lattice structure, showing a grid of nodes (circles) connected by lines. A central node is highlighted in red. Part (b) is a 3D visualization of the lattice structure, showing a dense network of colored lines (orange, yellow, green, blue, and red) representing the lattice paths.

(a) Grid Topology

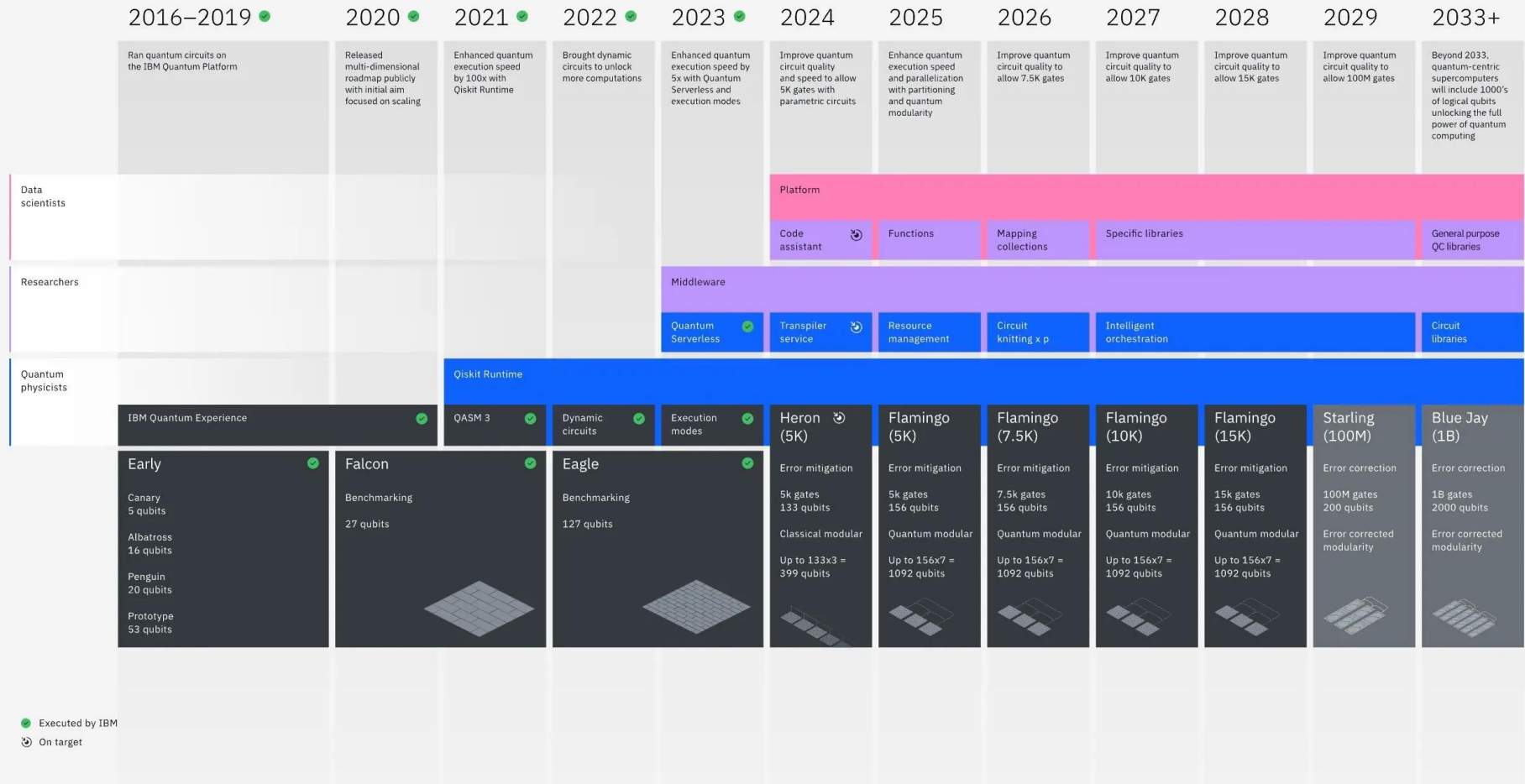
(b) IBM's Heavy-hexagonal

(c) Google's Sycamore

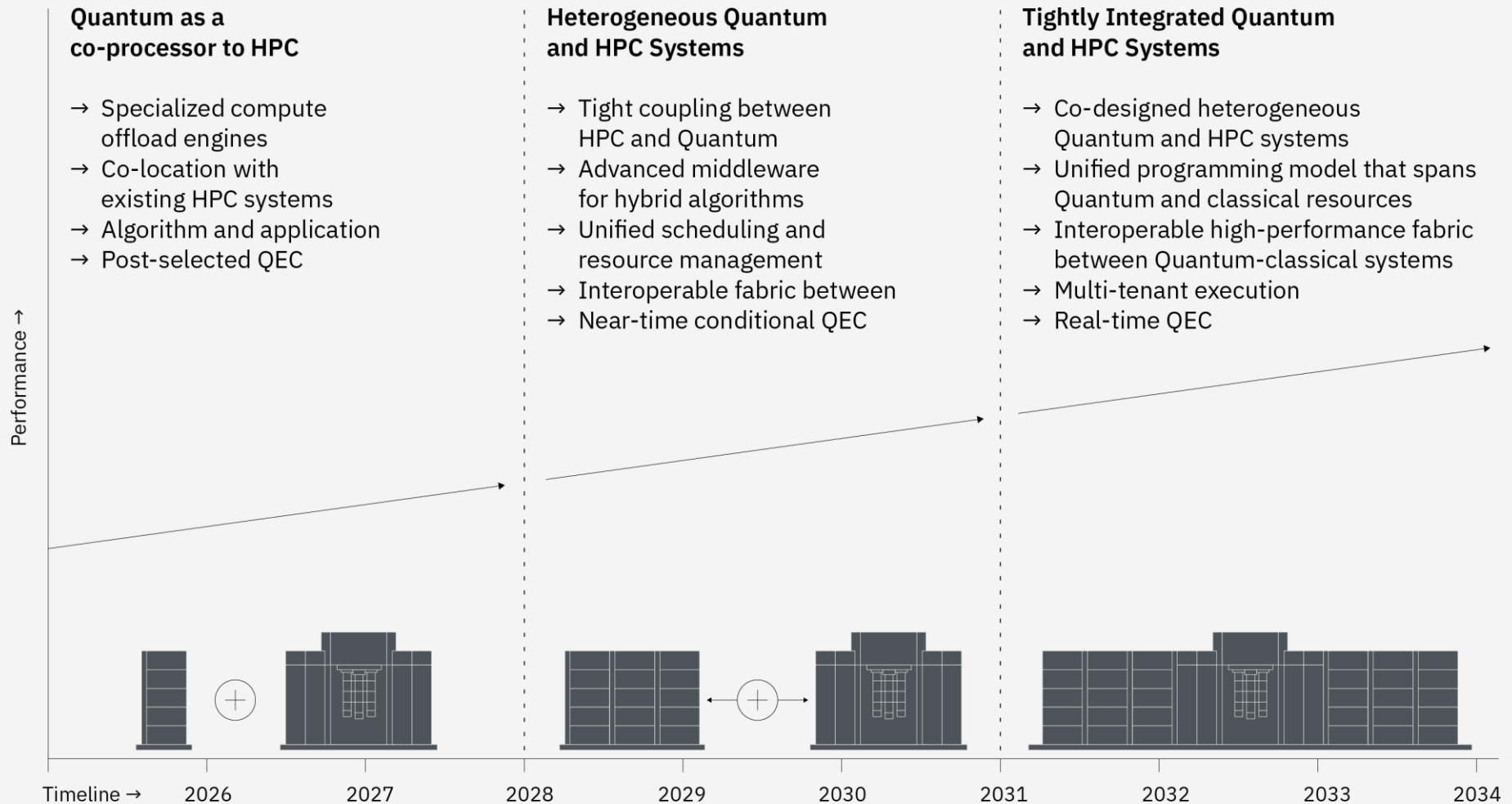
IBM's QPU Roadmap

Development Roadmap

IBM Quantum

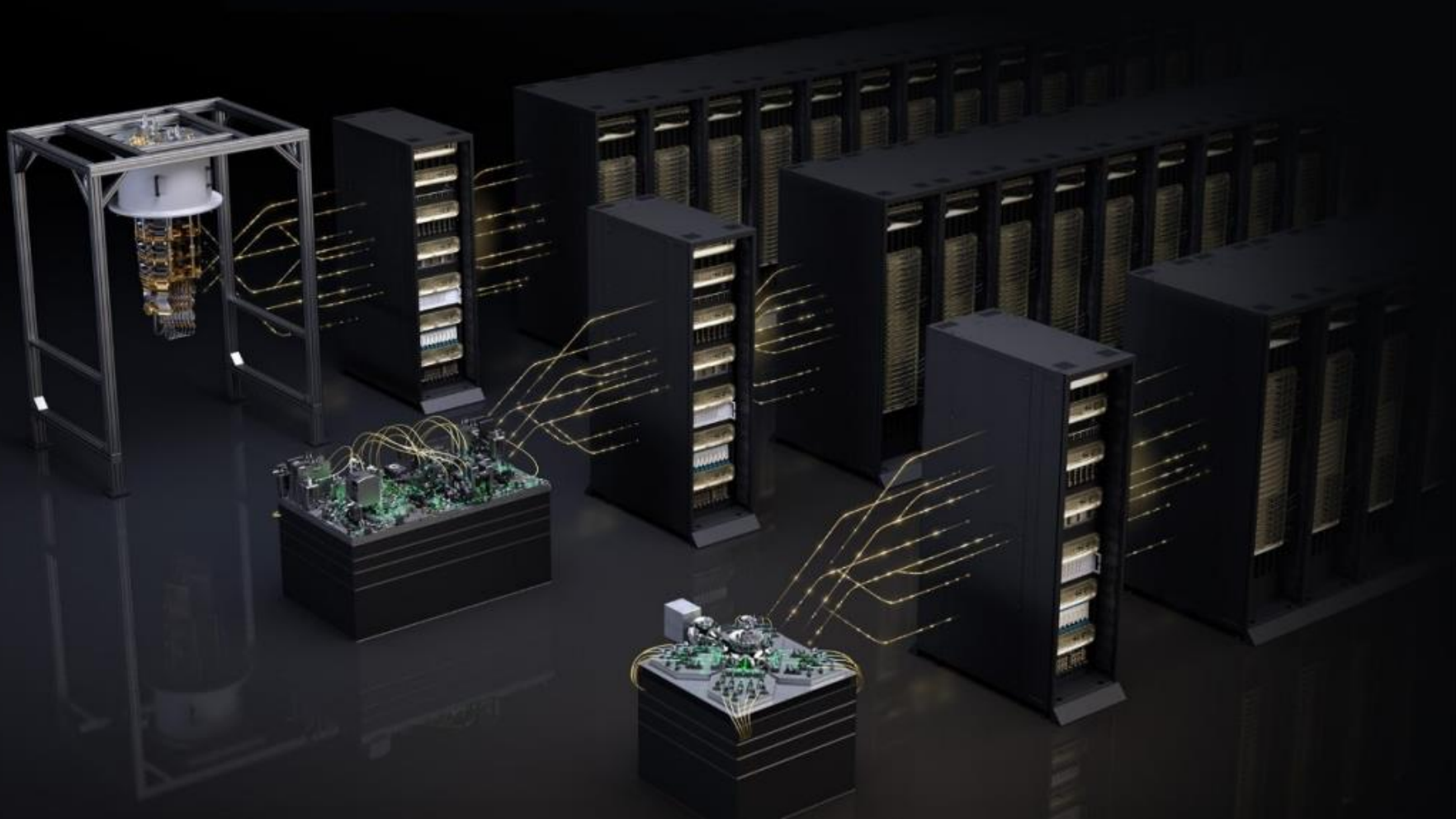


IBM's HPC+QC Roadmap

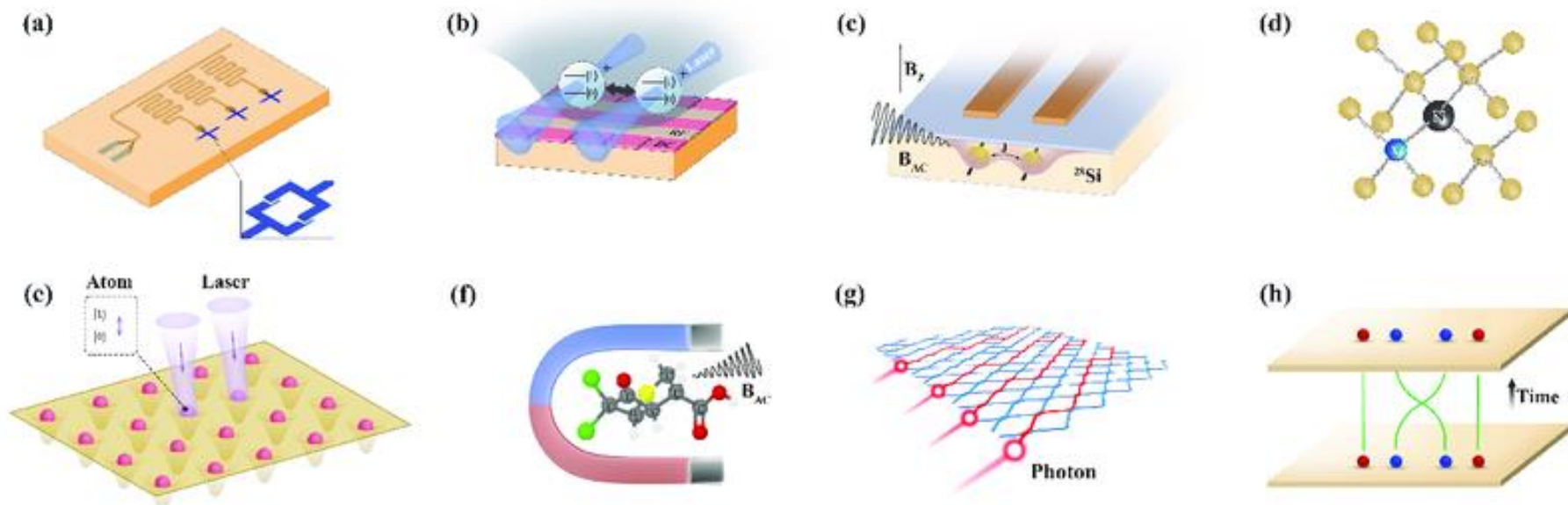


Nvidia NVQLink

- N Qubit * 7 Wire/Q * $\left(4 + \frac{N^2}{2}\right)$ Parameters(Self and interaction)
- $N = 1000 \rightarrow 35$ 億參數，GPU+AI 輔助



Different Quantum Platform



DiVincenzo's Criteria

- 可擴充、可量產(Scalable)
- 可初始化
- 足夠長的相干時間 (並能有足夠的邏輯電路深度)
- 能具備、執行通用邏輯閘
- 可針對個別位元進行獨立量測

Further for quantum communication:

- 可以本地量子位元和飛行量子位元(Flying Qubit)互相轉換
- 飛行量子位元可以準確的傳遞資訊

Jobs Opening

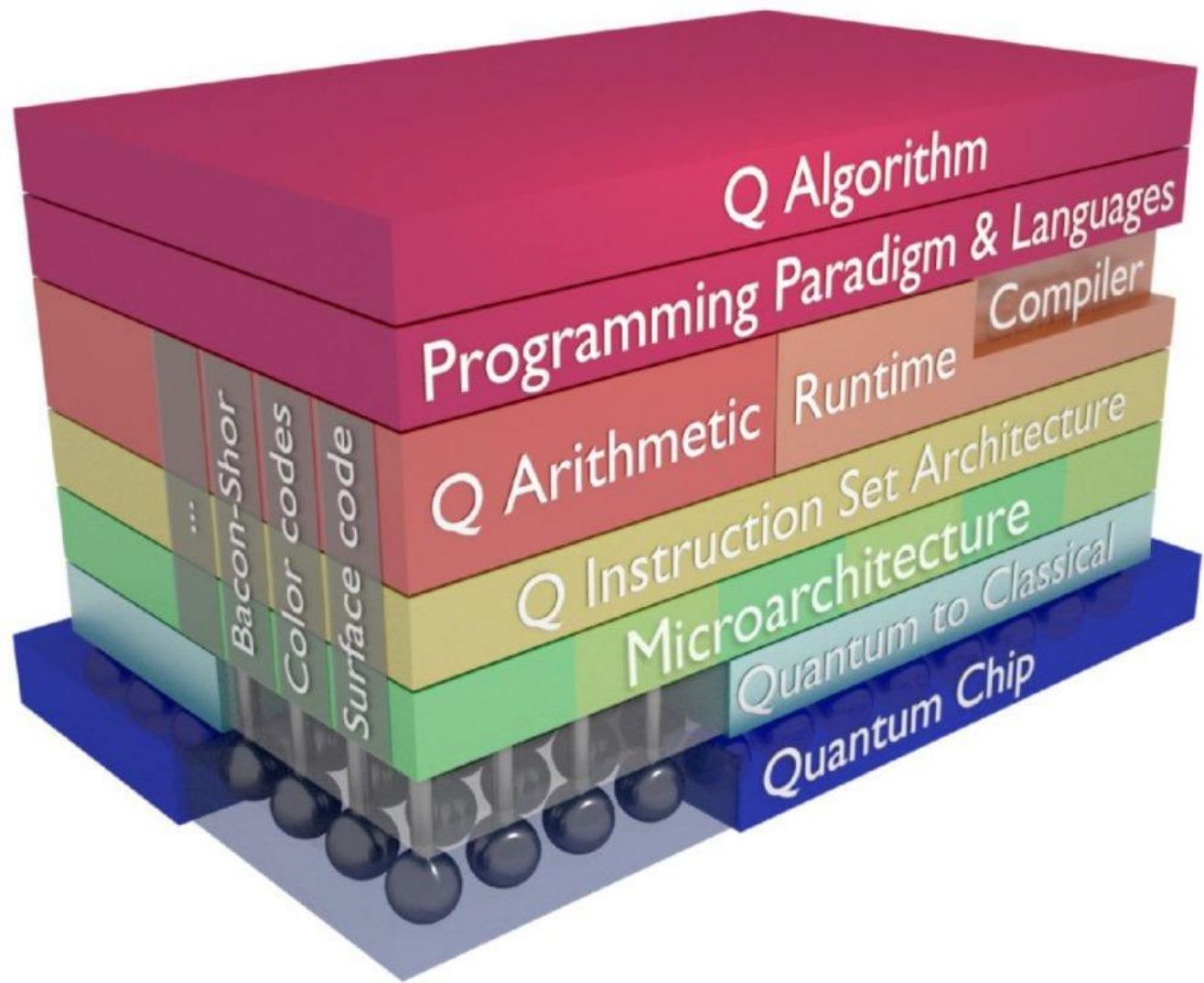
Quantum Technology Market Map – Quantum Computers

NON-EXHAUSTIVE, NO ORDER, EXCLUDES LABS

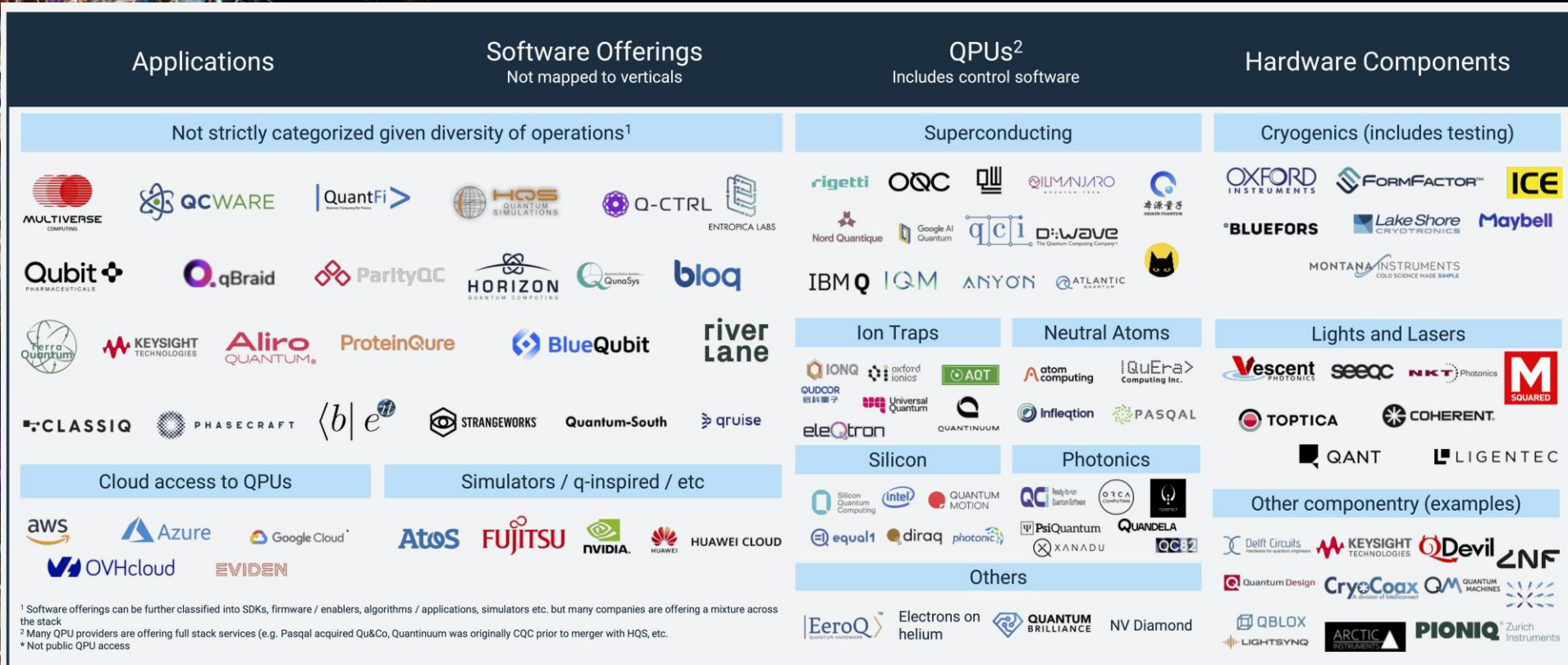


Source: The Quantum Insider Intelligence Platform

QC Architecture



More and More Jobs Opening





Thank you!